

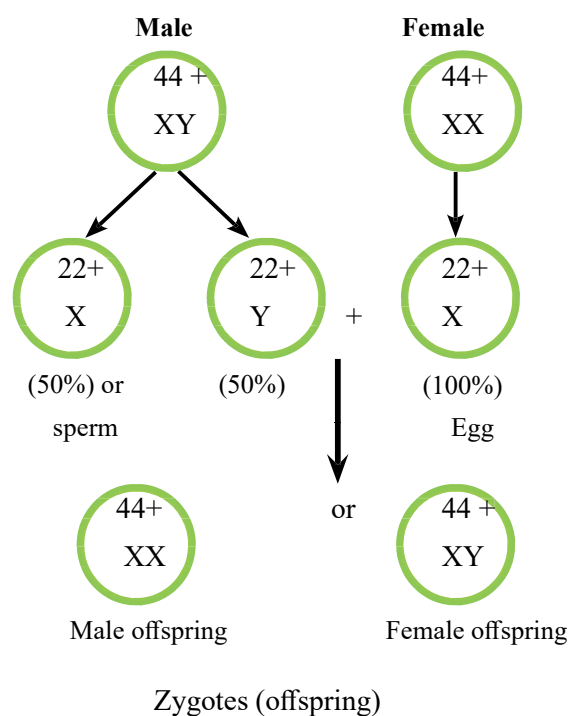


Figure 6.17: Sex determination of humans

Human sex linked characteristics

Certain characters of humans are carried on the genes located on the sex chromosomes. Those genes located on the sex chromosomes are called sex linked genes and the characters expressed by them are called sex linked characters.. Characters expressed by or carried on the X chromosome are called X- linked characters and the genes expressing or carrying those characters are called X- linked genes. On the other hand, characters which are expressed by or carried on the Y chromosome are called Y linked genes and the genes expressing or carrying those characters are called Y linked genes. Y chromosome carry only few genes other than those related to the sex. Some disorders carried on the Y-linked genes are transferred and expressed only through male progeny.

e.g. absence of certain Y-linked genes causes inability to produce normal sperm.

In addition to sex related characters, X chromosomes carry many other characters which are not relevant to individual sex.

e.g.

Red green colour blindness: *An X-linked recessive disorder characterized by the difficulty in perceiving differences between red and green colours.*

Haemophilia: An X-linked recessive disorder where one or more of the proteins required for blood clotting are absent. Haemophilic person run the risk of severe bleeding during injuries due to the delay in clot formation. (Campbell et al, 2015)

Inheritance of X-linked genes

The inheritance of sex linked characters or genes differ for male and female due to the XX genotype of female and the XY genotype of male. During the fertilization, X chromosome from both of their biological parents result a female zygote and on the other hand, X chromosome from the female parent and Y chromosome from the male parent results a male zygote. Thus, for females, X-linked recessive disorders are expressed only at their homozygous genotype. However in males, due to the presence of only one X chromosome, have only one recessive X-linked allele. Therefore, having a recessive X-linked allele with disorder is sufficient for expression.

Pleiotropy

In some occasions, expression of a single gene affects the expression of multiple traits which are not related to each other. The above phenomenon is called as Pleiotropy. Pleiotropic alleles are responsible for the multiple symptoms associated with certain hereditary diseases in humans, such as cystic fibrosis and sickle-cell disease.

Sickle-cell disease

Sickle cell disease is caused by an alteration in the haemoglobin protein of red blood cells. A single gene mutation is responsible for the above condition. In homozygous recessive individuals, all the haemoglobins are of the sickle-cell variety. People living in high altitudes or under physical stress experience low oxygen content in their blood. Low oxygen content in the blood may induce the sickle-cell haemoglobin proteins to get accumulated and results sickle shape in red blood cells. Sickle cells may clump and clog small blood vessels causing tissue and organ damage in several body parts. This may result renal failure, heart failure and thrombosis.

Cystic fibrosis

Cystic fibrosis is a disease condition causing thicker and stickier mucus than its normal nature. As a result, mucus get accumulated in the pancreas, lungs, digestive tract, and reproductive organs which cause lung infections, respiratory failure, poor digestion, and infertility.

The thickening of mucus is due to the excess chlorine secretion of defected chloride channels of the plasma membrane. The defect in the trans-membrane chloride channel occurs as a result of the Cystic Fibrosis Trans-membrane Regulator (CFTR) protein. The altered CFTR protein is due to the mutation of CFTR gene. This is identified as autosomal recessive disorder.

Epigenetics

Study of occurrence of certain phenotypes of certain characters controlled by factors other than their DNA sequence or genetic code is called epigenetics. This is due to ‘switching on’ and ‘switching off’ of certain genes by modifying nucleotides of a DNA sequence by methylation and demethylation, where methyl groups are added to wild type DNA sequence or else removed from a methylated DNA sequence. The above random occasions result different modified expression for a single DNA sequence.

Epigenetics results due to either inherited signals from parents or signals arising due to the environmental factors. Inheriting epigenetic traits from parents to the children's generation is called epigenetic inheritance. This may get reversed by various external stimuli from the environment. Some epigenetic influences result in inappropriate gene expressions leading to cancers.

Schizophrenia is a mental disorder that occurs due to the genetic defects. In some identical twins, only one of them gets schizophrenia and the other does not get it. This is due to two types of expressions for same DNA sequence, called epigenetics.

Population genetics

Hardy-Weinberg Equilibrium

Hardy-Weinberg Equilibrium is used to assess whether a population is evolving with respect to a particular characteristic/ genetic locus. The genetic makeup of a trait in a population would remain unchanged, if they are not evolving at that genetic locus. Therefore, the predicted data for a particular trait of a population can be compared with the actual data obtained from the same population. The comparison of both data as mentioned above may help to determine whether the population evolves or not for the considered trait.

Hardy-Weinberg Equilibrium Principle

In 1908 British mathematician **G.H. Hardy** and German physician **W. Weinberg**, independently showed that, in a population that is not evolving, allele and genotype frequencies will remain constant from generation to generation. This is now considered as key concept in population genetics and referred to as **Hardy-Weinberg Equilibrium Principle**.

To determine whether the allele and genotype frequencies have changed in consecutive generations, a Punnett square can be drawn, considering the combination of alleles in all possible crosses in a population.

The following example can be used to work out the Hardy Weinberg equilibrium.

A wild flower plant population showing incomplete dominance for the flower colour alleles have distinct phenotypes indicating their genotypes. i.e.

Plants homozygous for the C^R allele ($C^R C^R$) produce red pigment and have red flowers. Plants homozygous for the C^W allele ($C^W C^W$) have white flowers.

Heterozygous plants ($C^R C^W$) produce some red pigment and have pink flowers.

In the population of 500 flowers, there were 800 C^R alleles and 200 C^W alleles. Since the flower colour of the above plant type is determined by pair of alleles, 500 flowers may consist of 1000 alleles for their flower petal's pigmentation.

Therefore,

- allele frequency of C^R alleles (p) = $\frac{800}{1000} = 0.8$
- allele frequency of C^W alleles (q) = $\frac{200}{1000} = 0.2$

If the gametes are formed at random, the probability that an egg or sperm contains a C^R or C^W allele is equal to the frequency of each of these alleles in the population. Thus, each egg has an 80% chance of containing a C^R allele and a 20% chance of containing a C^W allele; the same is true for each sperm.

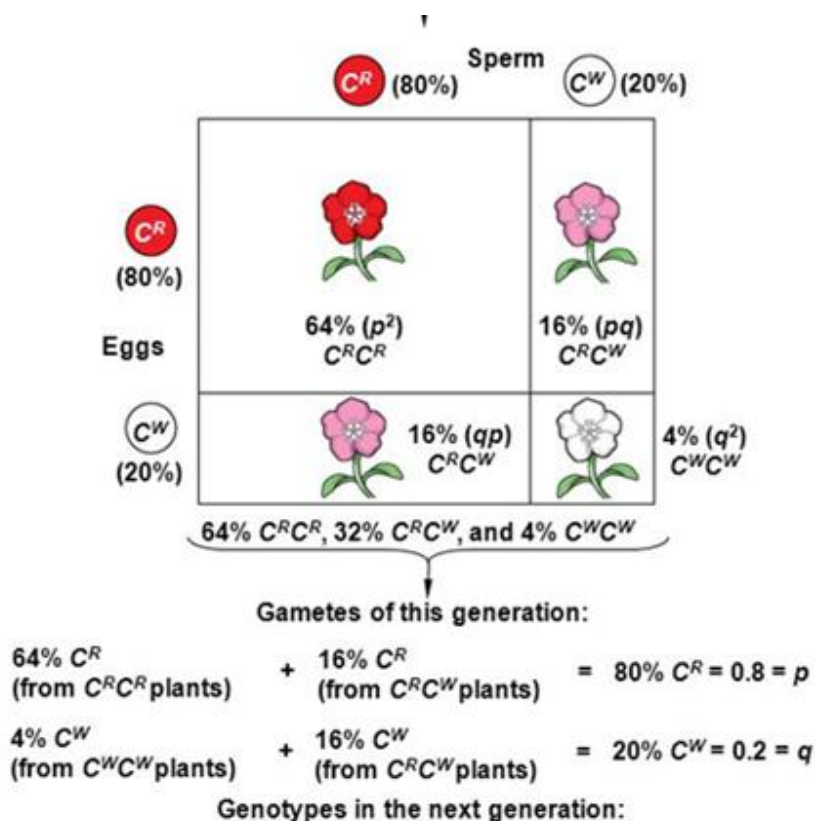


Figure 6.18: The Hardy-Weinberg Principle (Campbell et al, 2015)

During random fertilization, gametes fuse together randomly. Therefore, rule of multiplication can be applied to calculate the probability for each genotype combination.

According to Hardy-Weinberg equilibrium, if a character is determined by two alleles, the three genotypes will appear in the following proportions;

P_2 = frequency of dominant homozygotes

q_2 = frequency of the recessive homozygotes

$2pq$ = frequency of the heterozygotes

The probability that two C^R alleles will come together, $p \times p = p^2 = 0.8 \times 0.8 = 0.64$

Hence the proportion of $C^R C^R$ genotype in the progeny = 64%

The probability that two C^W alleles will come together, $q \times q = q^2 = 0.2 \times 0.2 = 0.04$

Hence the proportion of $C^W C^W$ genotype in the progeny = 4%.

$C^R C^W$ heterozygotes can arise in two different ways.

If the sperm provides the C^R allele and the egg provides the C^W allele,

the resulting $C^R C^W$ heterozygotes in the progeny, $p \times q = 0.8 \times 0.2 = 0.16 = 16\%$

If the egg provides the C^W allele and the sperm the C^R allele,

the resulting $C^R C^W$ heterozygotes in the progeny, $q \times p = 0.2 \times 0.8 = 0.16 = 16\%$

Thus, the total frequency of heterozygote in the progeny

$$pq + qp = 2pq = 0.16 + 0.16 = 0.32, \text{ or } 32\%$$

In the above example, only three kinds of genotypes are possible. When conditions suits for Hardy-Weinberg equilibrium, the sum of frequencies of all three genotypes equals 1. Thus, the equation for Hardy-Weinberg equilibrium can be written as below;

$$P^2 + 2pq + q^2 = 1$$

Conditions for Hardy-Weinberg Equilibrium

The Hardy-Weinberg approach describes a hypothetical population that is not evolving and fulfilling the following conditions.

1. **Absence of mutations.** Mutations result changes in alleles. Insertion, deletion, or substitution of nucleotides result altered alleles. This leads to modified gene pool.
2. **Occurrence of random mating.** Breeding occurs randomly without any influence that causes selectiveness. Mating of closely related individuals may alter the allele frequencies.
3. **Absence of natural selection.** All genotypes of the progeny are expected to survive irrespective to their differences, abilities and the environmental conditions. Variations in survival and reproduction of some genotypes may alter the frequency of alleles.
4. **Size of the population is extremely large.** In small populations, particular genotypes may disappear due to death or infertility. Therefore, larger the population, more likely to favor the Hardy-Weinberg equilibrium.

5. **Absence of immigration or emigration.** Individuals moving in and out of the populations may cause appearance of new genes and disappearance of existing genes. This is called gene flow and it may alter allele frequencies.

Naturally, most populations do deviate from Hardy-Weinberg equilibrium except for certain genetic loci. Slowly evolving populations may also do not deviate much from Hardy-Weinberg equilibrium and therefore, they remain as predicted for a non-evolving population.

Evolution and change in gene frequency

Evolution can be explained in terms of changes in allele (gene) frequencies over generations. A species evolves when changes in gene frequencies drive the species into a higher level of adaptation for a specific ecological niche.

Genetic variation within the population is the key to evolution. Mutation will create new alleles and migration will include it into the population to increase variation. Natural **selection will then choose the better adapted individuals based on their phenotypic variations, causing population to evolve.** As a result, after the evolution the population is at a higher adaptive level compared to the level of adaptation they showed before being evolved.

This concept can be illustrated with the evolution of the peppered moth in England during the time of industrialization. The moth had two phenotypic varieties based on their colour; dark and light. Prior to the industrialization of central England, the light-coloured allele was most prevalent. The light-coloured moths had an advantage over the dark coloured ones as they could hide on the white-barked trees to avoid predation from birds.

Due to the pollutants generated parallel to the industrialization, the light-coloured trees were stained dark. This exposed the light- coloured moths to predation, reducing their numbers. As a result, the light- colour allele became less prevalent. In its place, the dark-colour allele became more predominant, because dark moths could camouflage themselves better on the stained trees and avoid being eaten by their bird predators. The population evolved to a higher adaptive state with the change in gene frequencies (light – colour allele frequency went down while dark – colour allele frequency went up).

Plant and animal breeding

People have intervened in the reproduction and genetic make up of plants and animals since the dawn of agriculture, eight to ten thousand years ago. Early farmers selected the best looking plants and seeds and saved them to plant for the next season. Likewise, the best farm animals were allowed to mate with each other to preserve and improve their desirable traits.

This phenomenon wherein human beings interfere in the process of reproduction to allow only selective mating to occur, so that offspring with improved characters are produced is called **breeding** (as against natural reproduction).

With the science of genetics became better understood, plant and animal breeders used what they knew about the genes of a plant or an animal to select for specific desirable traits to develop improved plant varieties or animal breeds. The selection for features such as faster growth, higher yields, pest and disease resistance, larger seeds or sweeter fruits in crop plants, colour and pattern of the skin, hair or feathers in animals have now dramatically changed domesticated species compared to their wild relatives.

Importance of plant and animal breeding

In breeding programmes, the attributes, structure and composition of plant and animals are manipulated in such a way to make them more useful to humans. Accordingly, plant and animal breeding has a significant impact of world's agro-economy as discussed below;

Addressing world food and feed quality needs

An estimated 800 million people in the world, including 200 million children, suffer from malnutrition and associated health issues. Plant and animal breeding helps to enhance the value of food by improving their nutritional quality. For example, rice, which is the most widely eaten staple food, lacks many essential vitamins.

Another problem encountered in major food crops is the presence of toxic substances within them such as alkaloids in yam, cynogenic glucosides in cassava (manioc), trypsin inhibitors in pulses, and steroidal alkaloids in potatoes. Plant breeding is useful in reducing these toxic components and making them safer to eat. Plant breeding is also useful in making some plant products more digestible. For example, a high lignin content of the plant material reduces its value for animal feed which can be overcome with the use of breeding techniques.

Addressing food supply needs for a growing world population

It is anticipated that an additional three billion people will be added to the world population within the next three decades. Aligned with this population growth, an expansion in world food supply should be required to meet the projected needs. Unfortunately, land for farming is scarce and therefore more food will have to be produced on less land. This calls for improved and high yielding animal and plant varieties to be developed. In response plant breeding has produced super rice which has 50% more yield compared to the normal rice, super wheat which boost the

harvest by 20-40% of normal wheat and several high yielding corn, maize and soya bean varieties. The total production of meat and milk has also increased considerably over the years due to careful use of selective breeding techniques.

Need to adapt to environmental stresses

Weather and soil conditions can have a major impact crop yield. Climate changes and global warming are partly responsible for modifying the crop production environment (e.g., some regions of the world are getting drier and others saltier). To meet the increasing demand for food new cultivars need to be bred which can sustain these adverse conditions. For example, it is necessary to develop new plant types that can resist various biotic (diseases and insect pests) and other abiotic (e.g., salt, drought, heat, cold) stresses in the production environment. In response, genetically modified, pest resistant cotton, maize, and potatoes which carry Bt toxin, salinity tolerant rice varieties, cold tolerant tobacco, potato and strawberry varieties etc. are now available in agricultural industry. Likewise, both crop plants and farm animals (cattle, pig sheep goat etc.) with increased immunity to pathogens have also been produced through various breeding techniques.

Satisfying industrial and other end-use requirements

Consumers are having different requirements based on the texture, colour and composition of a particular food item irrespective of its taste or nutritional value. These diverse demands for the same food can be now successfully met through breeding procedures. For example, potato is a versatile crop used for food and industrial products. Different varieties are being developed by breeders for baking, cooking, fries (frozen), chipping, and for starch. These cultivars differ in size, specific gravity, and sugar content, among other properties. High sugar content is undesirable for frying or chipping because the sugar caramelizes under high heat to produce undesirable browning of fries and chips. Likewise, there is a high demand for seedless fruits such as grapes, melon and strawberries and also for leaner meat. Depending on these end-user requirements, it is possible to develop quality added products using animal and plant breeding techniques.

Developing animal and plant varieties with aesthetic values

Aesthetics is of major importance in horticulture as well as in the industry of ornamental and pet animals. The ornamental plant industry depends to a large extent on the development of new varieties that exhibit new flower/ leaf colours, varying sizes and attractive shapes etc. using plant breeding.

The pursuit of novelty has spurred a similar explosion of types in pet animals as well. Today selective breeding for numerous morphological features and functional abilities have given rise to nearly 400 dog breeds making them one of the most diversified species on earth. Likewise, there

are close to 50 rabbit breeds, vast number of bird varieties and an extensive range of ornamental fish.

Breeding techniques

Plant and animal breeders use numerous techniques to create new varieties with enhanced features. Many of these techniques have been successfully practiced over centuries even without the knowledge of the underlying genetics. Following section summarizes some of these traditional breeding techniques which has caused significant improvements in agriculture and farming.

Artificial selection

Artificial selection is the earliest form of biotechnology and has been used by humans for thousands of years. It, is a process of selective breeding, where plants or animals with specific traits were selected to breed so that their desired traits could be passed to the next generations to produce a high performing new variety. The method has made a huge impact on agriculture by way of improving plant and animal products before the discovery of more sophisticated technologies like genetic engineering.

The first prerequisite of artificial selection is the availability of variation with respect to the desirable characters. Once a population with a desirable variation is recognized, the best performing individuals for the desired feature are selected. For example, when selecting for fruit size in plants, only those giving the biggest fruits are chosen for the breeding programme and the rest of the population is discarded or rejected. The progeny of the selected individuals is grown further and again screened for the desired feature. This process is repeated sometimes for many generations, until a uniform plant population is attained which has the best-desired characters. Eventually, a new uniform crop variety with the desired characteristic is produced by this successive selection, followed by multiplication of the selected individuals.

The advantage of selective breeding is that it uses the processes of natural selection, but under direct supervision from carefully selected animals or plants with the desired traits. There are no genetic modifications or other forms of tampering that could potentially harm people and the risk to the plant or animal is often minimum.

Crops like corn and wheat are commonly selectively bred in order to obtain the highest yielding plants. Breeding animals with higher protein and lower fat percentages, as well as plants that have higher nutritional values, had been used to create food sources with a higher quality of nutrition. In addition, selective breeding has, effectively removed undesirable traits such as low resistance to disease, in some animals and plants.

However, selective breeding among animals can take a long time for the process to work. In horse breeding, for example, the given standard to establish a new breed is to have offspring with the desired traits to be produced over the course of 7 generations. This means it may take 25-50 years for the desired traits to become a foundational component of an animal.

Inbreeding and out breeding

Inbreeding

The breeding among genetically similar individuals are known as inbreeding.

Among plant breeders the term “inbreeding” is commonly used to mean self-fertilization, i.e. the fertilization of a flower with its own pollen or with pollen from a different flower on the same plant. This is done to produce an inbred variety, which is exactly the same generation after generation. Many important crops, such as wheat, oats, barley, and tobacco, are produced from seeds which are habitually self-fertilized.

However, in animal breeding the term " inbreeding " is used to refer to the mating of closely related individuals, as, for instance, the mating of father and daughter, brother and sister, or cousins. In both crop plants and farm animals, inbreeding brings uniformity of the required type while preserving the desired characters. Inbreeding is used for developing pure lines in agriculture as well as for research.

As a rule, inbreeding increases homozygosis and thus exposes harmful recessive genes which would have otherwise stay hidden among heterozygotes. Continued inbreeding, therefore, reduces genetic fitness of the population. As a result, the growth and fertility of the inbred population would go down with adverse effect on their productivity. Prevalence of genetic disorders might also increase among the inbred population. This phenomenon of having a reduced genetic fitness in a given population as a result of inbreeding is called **inbreeding depression**.

However, in agriculture and animal husbandry, positive effects of inbreeding will be harnessed as much as possible. To ensure this, only those offspring that are exhibiting the desired trait, without other negative ones, will be used for future breeding. The negatively affected individuals in the progeny are removed or are not be allowed to be bred. Thus, inbreeding is used in agriculture to help accumulation of superior genes.

Outbreeding

When plants or animals of different breeds (races) are mated with each other, it is known as **outbreeding** or **cross breeding**. This allows the desirable characters of the exotic parent, which the indigenous parent does not have, to be transmitted to the progeny.

For example, cross breeding is carried out by animal breeders to enhance milk and meat production. In India zebu breeds of cows and nondescript cows are crossed with exotic breeds like Holstein Fresian, Brown Swiss and Jersey bulls or their semen, to enhance the milk production potential of the progeny. Likewise crop plants like corn and hemp, are normally cross-fertilized.

Hybrid breeding

When genetically unrelated pure-bred plants or animals in the same species are mated with each other it is known as **hybridization** or **outcrossing**.

Generally, this is carried out with plants and animals who do not share common ancestors on either side of their pedigree up to four to six generations. The offspring of such a mating is known as the **out cross** and will possess stable characteristics and hybrid vigour. **Hybrid vigour**, also called **heterosis**, is the increase in such characteristics as size, growth rate, fertility, and yield in the hybrid organism over those of its parents.

Plant and animal breeders exploit heterosis by mating two different pure-bred lines that have certain desirable traits. The first-generation offspring generally show, in greater measure, the desired characteristics of both parents. This vigour may decrease, however, if the hybrids are mated together; so the parental lines must be maintained and crossed for each new crop or group desired.

In plant breeding, because creating hybrids involves many years of preparation to create pure lines that have to be constantly maintained so that F_1 hybrid seeds can be harvested each year, the seeds then become more expensive. Nevertheless, hybrid seeds have had a tremendous impact on agricultural productivity. Today, nearly all corn and 50% of all rice are hybrids. In the US, the widespread use of corn hybrids, coupled with improved cultural practices by farmers, has more than tripled corn grain yields over the past 50 years from an average of 35 bushels per acre in the 1930s to 115 bushels per acre in the 1990s. No other major crop anywhere in the world even comes close to equaling that sort of success story.

Many cultivars of popular vegetables or ornamental plants are hybrids. In terms of improved plant characteristics, tropical vegetable breeders can point to some rather clear achievements over the last two decades:

- **Yield improvement.** Hybrids often out yield traditional true breeding varieties (inbred varieties) by 50-100% due to its improved vigor, improved genetic disease resistance, improved fruit setting under stress, and higher female/male flower ratios.
- **Extended growing season.** Hybrids often mature up to 15 days earlier than local true breeding varieties. For many crops, the hybrid's relative advantage over the true breeding is most pronounced under stress conditions.
- **Quality improvement.** Hybrids have helped stabilize product quality at a higher, and more uniform level – this implies improved consumption quality (e.g. firm flesh of wax gourd, crispy taste of watermelon).

Interspecific breeding

In this approach, which is also known as interspecific hybridization, male and female organisms of two different species are mated. The progeny obtained from such a mating are usually different from both the parental species and may be fertile, partially fertile, or sterile.

Plants hybridize much more frequently and successfully than animals do. Pollen from flowering plants disperses widely and may land on flowers of other species allowing natural interspecific breeding to take place. Plant forms are less stringently controlled than animal forms, and so the intermediate form of a plant hybrid is more likely to be physiologically successful.

Often interspecific hybrids are sterile or for some other reason cannot interbreed with the parental species. Occasionally sterile interspecific hybrids can undergo a doubling of their chromosome set and become fertile tetraploids (four sets of chromosomes). For example, the bread wheats that humans use today are a result of two hybridizations each followed by chromosome doubling to produce fertile hexaploids (six sets of chromosomes). In such instances, the hybrids can become new species with characteristics different from either of the parents. Crop yields increase dramatically when hybridization is used to exceed one or more of the parents in size and reproductive potential. For example, boysenberries (*Rubus ursinus* x *idaeus*) were developed at Knott's Berry Farm in California. They are a result of a set of crosses between blackberries (*Rubus fruticosus*), European raspberries (*Rubus idaeus*) and loganberries (*Rubus* × *loganobaccus*).

For many fruit crop species, the use of interspecific breeding is increasing, in order to utilize naturally occurring sources of pest and disease resistance, fruit-quality components, etc. within the available germ plasm. For example, winter hardiness of apple was improved by making a hybrid species *Malus* × *domestica* from *Malus* × *asiatica* and *Malus pumifolia*.

However, among animals interspecific breeding is restricted to few species. Common examples include Mule (male donkey x female horse), Hinny (male horse x female donkey) and Liger (male lion x female tiger).

Note that in mules and hinnies, the common genus the parents belong to is *Equus* and in liger, its *Panthera*. Other examples are zebra/donkey cross resulting in an offspring called zonkey, zebra/horse cross resulting in zorse. The offspring from this cross could develop into adults, but may not develop functional gametes. Sterility is often attributed to the different number of chromosomes the two species have, for example, donkeys have 62 chromosomes, horses have 64 chromosomes.

Genetic principles of breeding techniques

From the beginning of plant and animal breeding, farmers made use of principles of genetics, with or without awareness of these concepts. Following section explains, three most widely used genetic principles applied in animal and plant breeding at present.

Polyploidy

Polyploidy refers to the presence of more than two complete sets of homologous chromosomes per cell nucleus. This is a widely used principle in plant breeding. In plants, polyploidy can be induced artificially using antimitotic agent, colchicine.

One of the most important consequences of polyploidy for plant breeding are the increment in plant organs (“gigas” effect) caused by the larger number of gene copies. Polyploid individuals may thus exhibit larger organs compared to their diploid counterparts, such as roots, leaves, tubercles, fruits, flowers and seeds. Polyploid plants also have lower growth rates, and tend to flower later or over a longer period of time than related diploids, which is a desirable feature for ornamental breeding.

In addition, polyploidy often results in reduced fertility due to meiotic errors, allowing the production of seedless varieties such as the triploid watermelon. On the other hand, when the crossing between two species is not possible because of differences in ploidy level, polyploids can be used as a bridge for gene transferring between them. Similarly, the genome doubling in a newly formed sterile hybrid allows the restoration of its fertility.

Genome redundancy (having additional gene copies due to increase ploidy) have other benefits as well. It promotes a “buffering” effect in which the deleterious alleles are masked by the extra copies of wild-type alleles. At the same time, it allows functional diversification of redundant gene copies, in which one member of a duplicated gene pair mutates and acquires a novel function, without compromising essential functions.

The increment in heterozygosity is another feature that accompanies polyploidy. Higher levels of heterozygosity have been positively related to vigor increment in maize, potato and alfalfa improving the product quality and increasing the tolerance to both biotic and abiotic stresses.

Mutation Breeding

Methods for inducing mutation have the potential of producing new sources of genetic variability for crop breeding. These methods can be employed when it appears there is little, or no, variability for the character to be improved available within the gene pool of the species. This method of inducing desirable mutations in crop plants using either chemical or physical agents was termed **mutation breeding**.

Several agents can be used to cause mutations. This include ionizing radiation such as gamma rays, protons, neutrons, alpha and beta particles and chemicals such as sodium azide and ethyl methanesulphonate. Since the desirable mutations induced by these treatments are found at a very low frequency (0.1% of total mutations), breeders have to screen a large population to select a desirable mutation. In addition, most mutations act in a recessive fashion and are likely to be masked by their dominant allelic counterparts making the screening procedure even harder.

The effectiveness of using induced mutation depends on the breeding system of the plant. Its use in self-pollinated plants is likely to be more successful than in crosspollinated ones. Populations of cross-pollinated plants usually possess stores of genetic variability in the recessive condition and it would not be likely that induced mutation would produce significant amounts of new variability. Further, induced mutation is potentially useful in the improvement of asexually propagated crop plants.

Despite these limitations, mutation breeding efforts continue around the world today. It has improved both morphological and physiological characteristic of both crop and ornamental plants such as flower colours, seed size, crop yield, disease resistance and salinity tolerance, drought tolerance and early maturity. Examples of plants that have been produced via mutation breeding include wheat, barley, rice, potatoes, soybeans, and onions.

Genetic modification

Genetic modification, also called **genetic engineering**, is the direct manipulation of an organism's genes to change the genetic makeup of cells. In this method, genetic material is obtained from one organism showing a desired trait and will be inserted to another second organism using recombinant DNA technology, so that the receiving organism will also show the same desirable trait. Thus, by transferring genes within and across species, improved or novel organisms are produced.

In traditional plant breeding techniques transfer of genes is limited to the closely related species or genera. For example, traditional breeding techniques could not be used to insert a desired gene from daffodil into rice because the many intermediate species between rice and daffodil and their common ancestor is extinct. With genetic engineering, however, such gene transfers can be done more quickly, more specifically, and without the need for intermediate species. The term **transgenic** or **genetically modified organism (GMO)** is used to describe organisms that have been engineered to express a gene from another species. Advocates for plant biotechnology believe that the genetic engineering of crop plants is the key to overcoming some of the most pressing problems of the 21st century, including world hunger and fossil fuel dependency. Examples of transgenic plant varieties include transgenic papaya that is resistant to ring spot virus, “golden rice,” with increased levels of beta-carotene and salinity resistant rice varieties among others.

Natural or artificial breeding: Advantages and disadvantages

Although artificial breeding is extensively practiced today with remarkable economic gains, there are several drawbacks in the method in comparison to the natural breeding.

The artificial breeding is geared towards producing a uniform set of plants or animals with traits desirable to humans. The development of this uniformity needs compromising the variability within the species. This reduction of genetic diversity will have adverse consequences on the evolutionary fitness of the species leading to low resistance to infections, higher prevalence of congenital anomalies and reduced fertility. For example, when a population of animals or plants with the same genetic traits are attacked by a pathogen to which they are susceptible, the entire population is likely to suffer due to the absence of the resistant trait within the gene pool. This limited opportunities for the natural selection to act upon the population is likely to drive them towards a lower fitness plateau.

On the other hand, natural breeding can rule out weaknesses and disabilities which affect survival by allowing natural selection to act upon the species. This will produce fitter and stronger individuals in the long run. However, the natural selection would not guarantee a productivity increase in the consumer perspective, despite the increase in genetic fitness.

As discussed earlier, sometimes inbreeding is practiced as an artificial breeding technique. This would result in an increase in homozygosity which would in turn increase the expression of recessive deleterious mutations that would otherwise stay masked within heterozygotes. This can cause the population to undergo inbreeding depression with adverse effects on the overall fitness.

Sometimes artificial breeding can exhibit negative correlated responses. This refers to the fact that while artificial breeding is improving certain characters in the population, simultaneously and unintendedly it could lead towards deterioration of other characters that are not under direct

observation. For example, the shape of the skull in some dog breeds has made it difficult for them to eat normal food because of the upper jaw being much shorter than the lower jaw, such as in the case of Boxer or the Bulldog. Likewise, selection for large offspring has resulted in a high fraction of difficult births, sometimes requiring caesarean sections in the Texel sheep, and even almost as a standard way of delivering in the beef cattle breeds Belgian White-and-Blue cattle and the Dutch Improved Red-and-White. This kind of negative responses are difficult to predict in advance and usually only visible after the new breed is established.

Although there are negative impacts artificial breeding is still preferred over natural breeding for the numerous advantages it could bestow upon overall animal and plant productivity as discussed earlier.

Notes:

This is to acknowledge that some of the diagrams used in this book have been taken from various electronic sources using internet . This book is not published to make profit and sold only to cover cost.

The resource book is prepared according to the subject content and learning outcomes of the G.C.E. (A.L) Biology new syllabus which is implemented from 2017.

The content of this Resource book declares the limitation of the G.C.E. (A.L) Biology new syllabus which is implemented from 2017.

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07

Molecular Biology and Recombinant DNA Technology

Nucleic acids are capable in directing their own replication from their monomers. In most organisms, DNA is found as the genetic material. However, some viruses such as influenza virus, consist RNA as their genetic material. Accurate replication of DNA, its transmission from one generation to the other and its ability to store and express hereditary information facilitates DNA for acting as vital genetic material in organisms.

DNA Double Helix Model: James Watson and Francis Crick, based on the X-ray crystallography of a DNA molecule obtained by Rosalind Franklin, proposed the double helix model to describe how the six molecules; de-oxy ribose sugar, phosphate group and four different nitrogenous bases, are organized into DNA and explain its properties. According to this model, DNA is a twisted ladder (spiral staircase) in which the rails are composed of alternating phosphate and sugar molecules forming the back bone. The steps of the ladder are paired nitrogenous bases. A purine is paired with a pyrimidine as per the base pairing rule with either two hydrogen bonds (A=T) or three (G $\equiv$ C). T.H. Morgan and his group, through their experiments concluded that chromosomes are made up of DNA and proteins and genes as regions of chromosomes.

Architecture of Chromosomes

Architecture of chromosomes is the way the DNA molecules are arranged in the nucleoid or nuclear region of the cytoplasm of the prokaryotic cells or in the nucleus of eukaryotic cells.

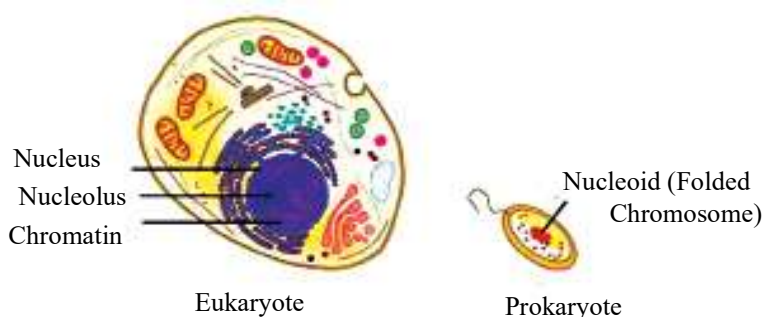


Figure 7.1: DNA is packaged into a nucleus in eukaryotes and a nucleoid in prokaryotes.

DNA of both prokaryotes and eukaryotes are called chromosomes. However, true chromosomes are present only in eukaryotes.

Prokaryotic (bacterial) chromosome is a single double stranded circular DNA molecule associated with a few protein molecules. Eukaryotes have several chromosomes, each consisting of a single double stranded linear DNA associated with histones and other protein molecules.

When the size of all chromosomes of an organism are considered, it is an enormous amount of DNA. These facts illustrate that, a cell has a huge problem in having its DNA either in the nucleoid of prokaryotes or the nucleus of eukaryotes. The containment of the genome/DNA in the nucleoid or in the nucleus is called DNA packaging.

Packaging of DNA in prokaryotes is facilitated by the proteins associated with the DNA molecule. The protein molecules cause the DNA to coil (or fold or loop) and supercoil, so the DNA molecule is compacted into the nucleoid. The DNA molecule is initially coiled into loops and these loops then independently supercoil into domains identifiable in electron micrographs. The loops of compacted mass of DNA are bound to a 'core' consisting of RNA and protein. The 'core' also attaches the chromosome to the membrane. The supercoiled DNA can be relaxed by introducing single strand nicks. Since the chromosomes are attached to the membrane and held together by the 'core' acting as a barrier for rotation, the domains can relax and supercoil independently. This is important in the transcription of specific genes. Removal of RNA will lead to loss of independence of the loops.

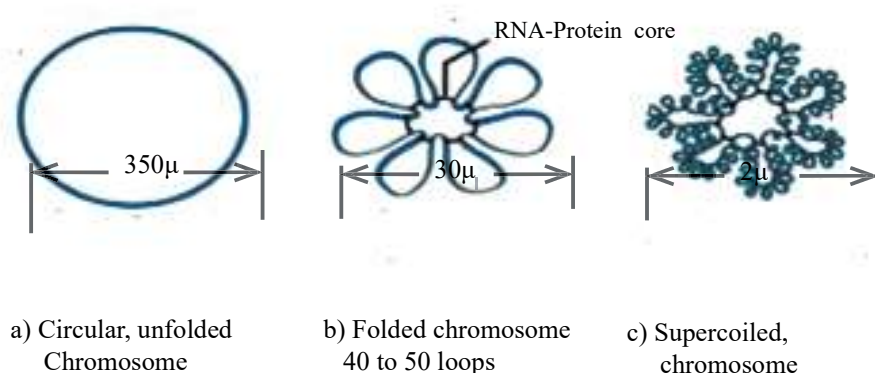


Figure 7.2: Folding and supercoiling compact the prokaryotic chromosomes

In addition to the chromosomal DNA, certain prokaryotes have extra chromosomal genetic elements called plasmids. They are also circular DNA and are coiled and supercoiled.

Eukaryotic chromosomes are associated with a large number of proteins called histones that help to organize the DNA inside the nucleus. This DNA-protein complex is known as chromatin. The chromatin may be lightly packed as in euchromatin or tightly packed as in heterochromatin. Euchromatin is rich in genes and is probably active in transcription. Heterochromatin consists of nucleotide sequences which are mostly inactive. They may contribute to gene regulation, epigenetic inheritance and protection of chromosomal integrity.

In the first level, the double helix winds around a complex of eight histone molecules. These are called nucleosomes and they look like beads of a necklace. The adjoining beads of nucleosomes are linked together by a stretch of DNA: linker DNA. (Figure 7.3)

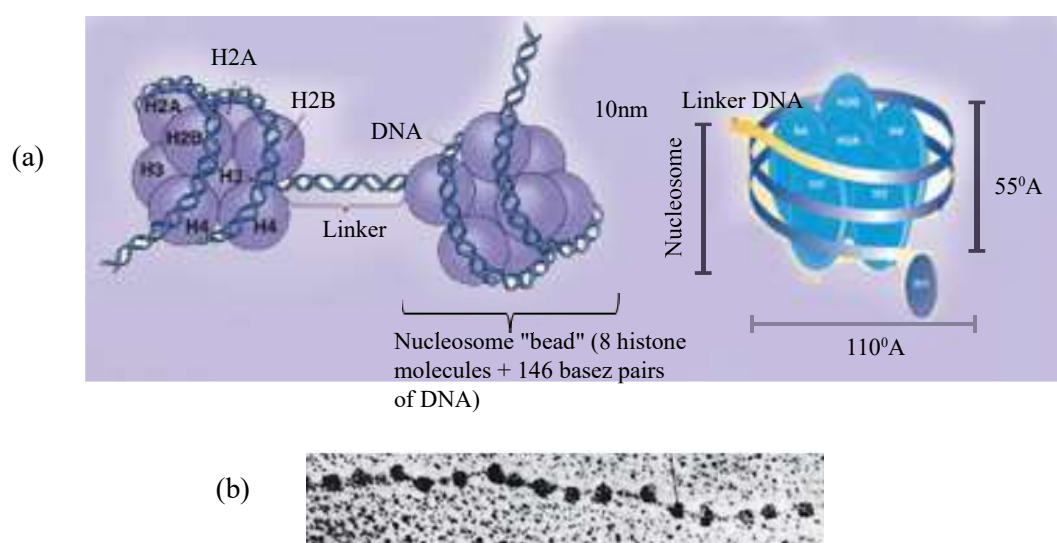


Figure 7.3: a) First level of packing: Forming beads of nucleosomes joined together by linker DNA, b) Electron micrograph showing nucleosome (beads) and linkers (strings)

In the second level, the nucleosomes twist and pack in a spiral fashion to form a chromatin fibre of roughly 30 nm in diameter: 30 nm fibres from 10 nm fibres. (Figure 7.4)

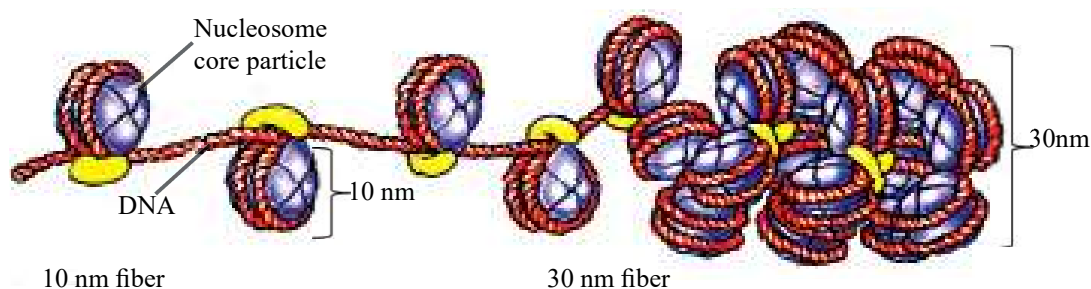


Figure 7.4: Formation of a 30 nm fiber (Cylindrically coiled)

At the third level, the 30 nm fiber forms loops, called looped domains, attached to a protein scaffold. This structure has a thickness of 300 nm. (Figure 7.5)

Finally, at the fourth level, the looped domains coil, fold and further compact to form the mitotic chromosome. The diameter of a chromatid is about 700 nm. In the metaphase chromosome, the chromatids are already replicated (Figure 7.5).

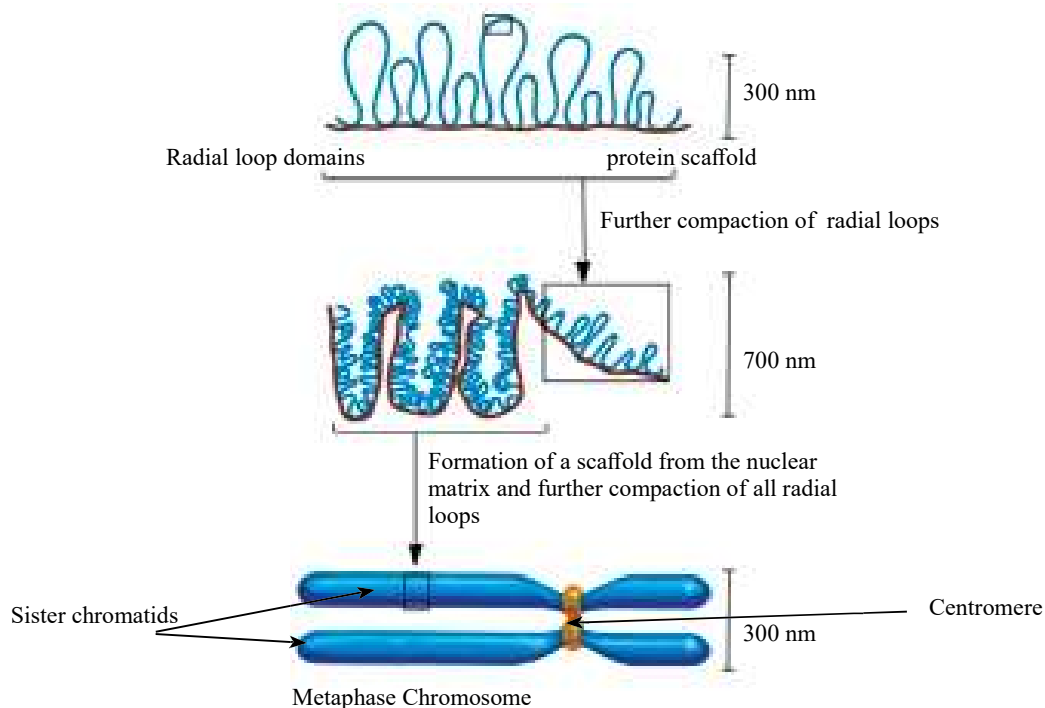


Figure 7.5: Compaction into looped domains on scaffold and into chromatids



Figure 7.6: Metaphase chromosomes (discrete units) and chromatin in the interphase

DNA Replication -

This is the process which copies a double stranded DNA molecule to produce two identical copies. Because, DNA replication process is basically similar in prokaryotes and eukaryotes. However, there are differences in enzymes involved. This is because the Eukaryotic nuclear DNA is organized into chromosomes and contain histones in their structure for packaging, and the prokaryotic DNA are usually found as circular molecules, which are supercoiled for packaging.

Importance of DNA replication

- The essential information for life is stored in DNA. Therefore, the new cells produced must receive the DNA from their parent cells as, each cell in the body of a diploid organism should contain the same genetic information as was in the zygote. A multicellular organism grows by addition of new cells.
- Damaged or dead cells are also replaced by new cells.
- In asexual reproduction, the offspring is identical to the parent cell. This is possible only because DNA has the ability to replicate and provide an identical set of genetic information stored in DNA to each daughter cell by mitosis.
- In sexually reproducing organisms, meiosis occurs at some point of life cycle in order to maintain a constant number of chromosomes. DNA replication occurs prior to meiosis.
- DNA replication is a very accurate process, so that it makes identical copies. However, rare errors occur in DNA replication, introducing mutations which results in variation. Variation leads to evolution of organisms.
- Therefore, DNA replication is important in maintenance of the life of an individual organism and continuation of a species.

The process of DNA Replication in prokaryotes

The entire replication process is carefully controlled and coordinated by a number of enzymes and other proteins.

DNA is synthesized on the existing DNA strands of the double helix of the DNA molecule. The parent DNA strands are used as the templates. Thus, the newly synthesized DNA double helix contains one parent DNA strand and a new complementary strand.

First of all, the tightly packed DNA (supercoiled DNA in prokaryotes and chromatin in eukaryotes) has to relax, so that the DNA replication machinery can have access to the place where the replication begins.

The separation of the double helix occurs at the origin of replication. Origin of replication or 'Ori' is a specific sequence of DNA to which proteins that initiate DNA replication bind. Starting from this, the entire circular DNA is replicated on both directions. Since the enzyme synthesizing the new DNA strand can move only in one direction (5' to 3' direction) one of the new strands can be synthesized continuously while the other will be synthesized as small fragments. The

small fragments of lagging strand are called okazaki fragments. These are called leading and lagging strands respectively. Replication of large DNA molecules may begin at several origins of replication to speed up the process.

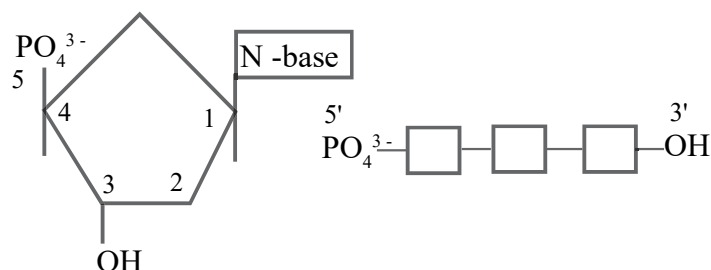


Figure 7.7: 5'- phosphate and 3'- OH of a DNA Molecule

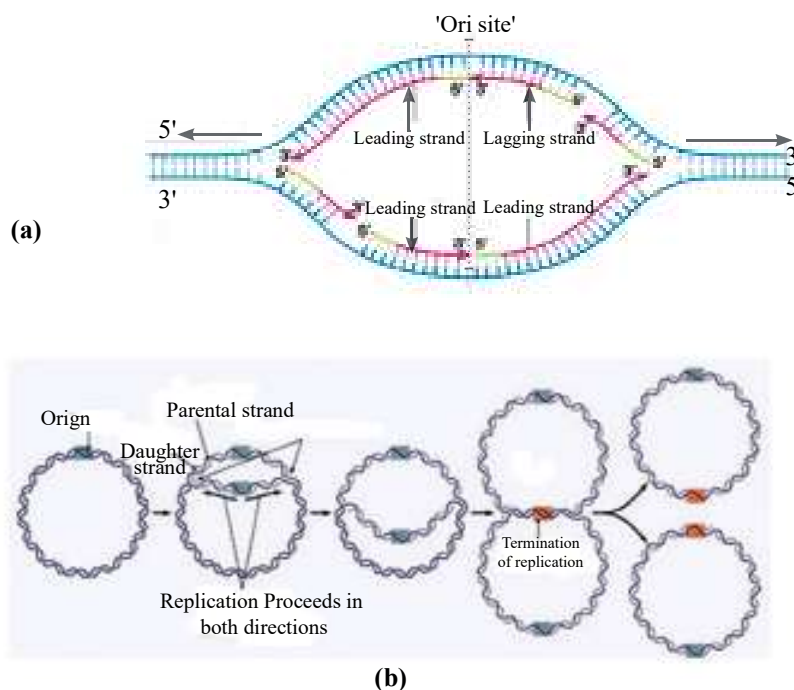


Figure 7.8: (a) Details of DNA replication (b) DNA replication of small circular DNA

Functions of major enzymes and other proteins of replication machinery

A number of enzymes and other proteins are required in the replication of DNA. These proteins assemble at the origin of replication. The major enzymes involved in the DNA replication are: helicase, topoisomerase, primase, DNA polymerases and DNA ligase. There are several other proteins in the replication machinery including Single Strand Binding (SSB) proteins.

Helicase: These are the enzymes that unwind the double helix and separate the two strands of a DNA molecule, with the expense of energy as ATP. The separation occurs by breaking hydrogen bonds between the base pairs of the two strands. This is important for exposing the two strands to function as template for new DNA synthesis (replication).

Topoisomerase: These enzymes are working ahead of the direction of DNA synthesis. The untwisting of the strands at one place causes further twisting and strain on other places. Topoisomerase enzymes, introduce breaks on one or both strands of DNA, twist the molecule to relieve the strain and then reseal the cut ends.

Single Strand Binding Proteins (SSB): These protein molecules bind to exposed single stranded DNA to prevent re-pairing of the separated DNA strands and to stabilize them. If the two strands re-pair, then they cannot serve as templates for new DNA synthesis.

Primase: In order to synthesize a new DNA strand on a template, complementary deoxyribonucleotides should be added one after the other in the correct sequence. This is done by DNA polymerase, but DNA polymerases can add nucleotides only to the 3' end of an already formed nucleic acid chain. For this, a small fragment of nucleic acid chain is sufficient and this fragment is called a primer. Primase is a type of RNA polymerase which can initiate synthesis of RNA on a DNA template by adding ribonucleotides. Primase adds a short RNA primer on the DNA template forming DNA-RNA hybrid to facilitate the action of DNA polymerase (figure 7.9).

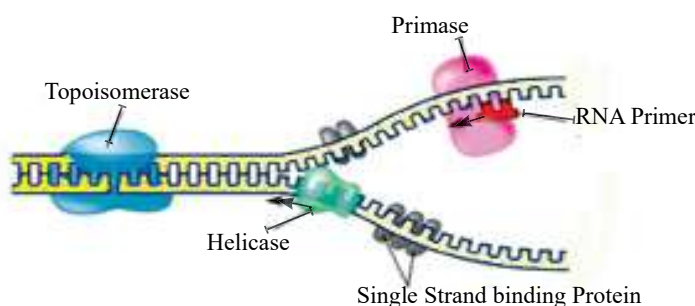


Figure 7.9 : ds DNA replication at the origin of replication forming replication bubble

DNA polymerases: There are several types of DNA polymerases. One type of DNA polymerase initiates DNA polymerization by adding a deoxyribonucleotide to the 3' end of a primer and continues polymerization on 5' to 3' direction elongating the new DNA strand with deoxyribonucleotides with bases complementary to template (figure 7.10).

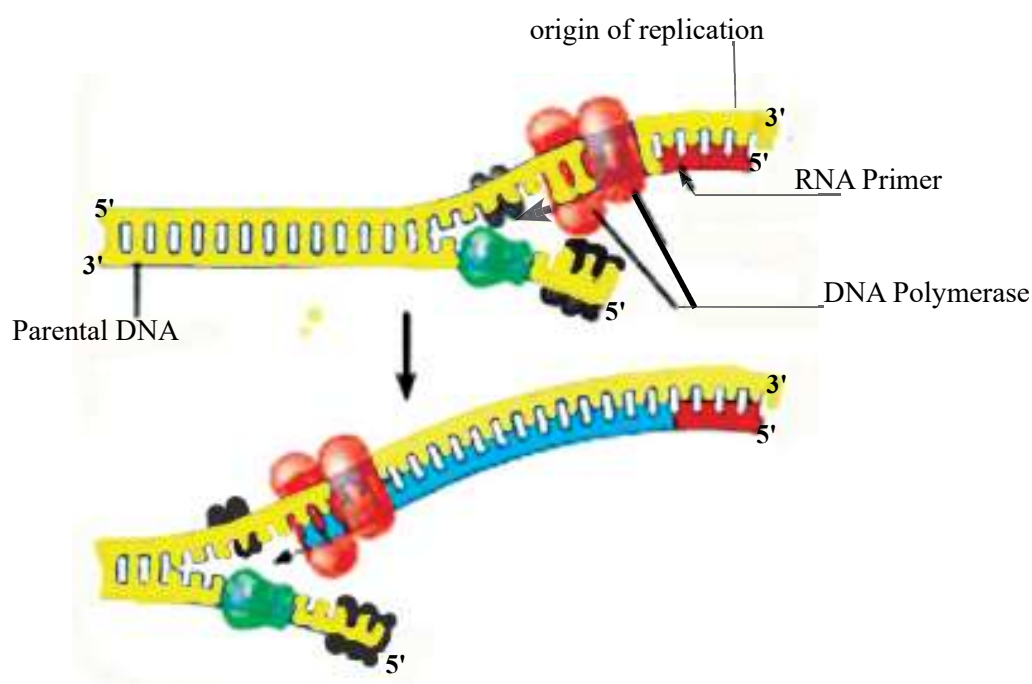


Figure 7.10: DNA Polymerase extends the new strand of DNA starting from 3' end of RNA primer

This DNA polymerase is almost 100% error-free in adding correct complementary nucleotides to growing strand, according to the nucleotide sequence of the parent DNA strands. However one error in 10^5 added nucleotides is possible, has a proofreading mechanism and therefore can correct its own mistakes and the error rate will be reduced by 100,000 times to one in 10^{10} . Therefore, the daughter DNA molecules are almost identical to the parent DNA molecule, and of course to each other.

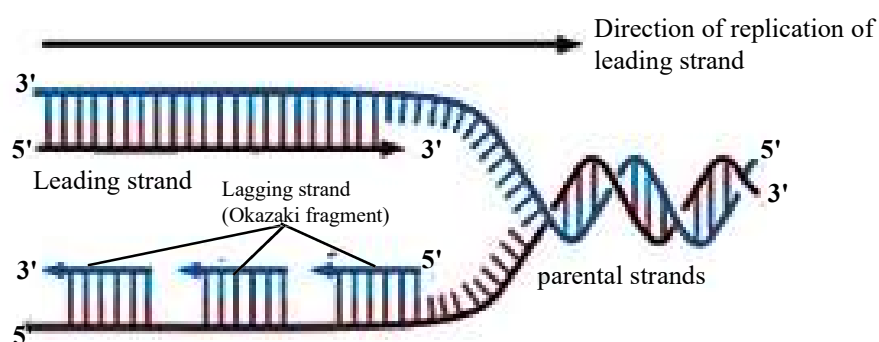


Figure 7.11: How DNA Polymerase solves the problem of antiparallel nature of DNA molecule

When a wrong nucleotide is added to the growing chain of DNA by DNA Polymerase this mismatch is identified by the same enzyme, stops adding the next nucleotide, removes the incorrect nucleotide exonuclease activity and continues again its polymerase activity. This is called as proofreading activity of DNA polymerase.

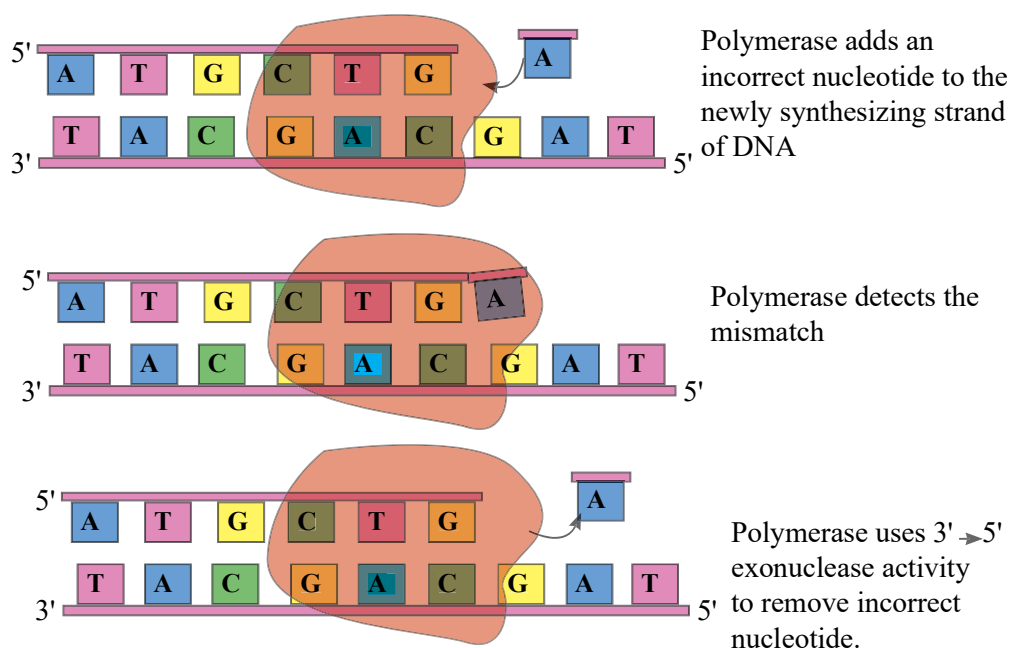


Figure 7.12. Proofreading activity of DNA Polymerase

Another type of DNA polymerase identifies DNA-RNA hybrid, removes ribonucleotides and replaces those with deoxyribonucleotides and so replaces the RNA primer with DNA. Now the DNA segment is incomplete, but the DNA polymerase unable to correct the ends of okazaki fragments, results gaps.

DNA ligase: In DNA synthesis, it joins the newly synthesized adjacent fragments making a complete strand by the formation of a phosphodiester bond. It seals the gaps of the newly synthesized DNA strand.

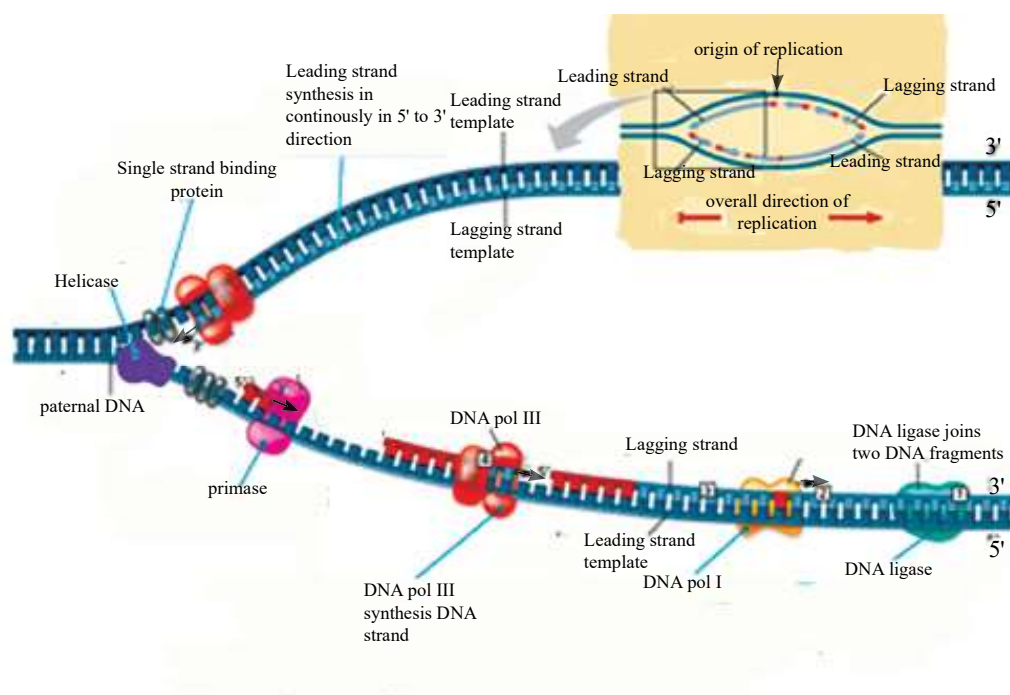


Figure 7.13: Overall process of DNA replication

Overall Process of DNA Replication

- Relaxation of tightly wound DNA
- Unwinding/ unzipping of the double helix
- Stabilization of single stranded DNA
- Priming of DNA synthesis by a RNA primer
- Extension of new DNA strand - Leading strand: Continuous
 - Lagging strand – Discontinuous
- Removal of RNA primer and replacing RNA (ribonucleotides) with DNA (deoxyribonucleotides)
- Sealing gap between the adjacent nucleotides

Similarities and Differences between Prokaryotic and Eukaryotic DNA Replication:

Eukaryotes share many characteristics of DNA replication of the Prokaryotes. They also use helicase to unwind the dsDNA, and use DNA polymerases for the polymerization reaction. The replication is originated in both types at specific sequences (origin of replication – Ori) to initiate DNA replication, packed DNA is relaxed with topoisomerase and replication process move in the same manner and therefore having leading and lagging strand. RNA primers are formed and replaced. The gaps are sealed by ligase.

Although the process is superficially similar, there are several significant differences too. The size of the DNA molecule in a eukaryotic chromosome is much larger than that of a circular DNA molecule of a bacterium. Therefore prokaryotes usually have one Ori while an Eukaryotic chromosome has several Ori. The DNA polymerases in eukaryotes and prokaryotes are different from each other in their structure, while having the same functions. The DNA replication of the prokaryotes occurs continuously, whereas in eukaryotes it happens only in the S-phase of cell cycle.

DNA repair and its significance

Damage to DNA can be caused by various chemical and physical agents introducing mismatches in DNA double helix and may lead to a permanent change in the sequence of DNA. This may also be caused by errors in DNA replication that goes unnoticed by proofreading. This is called mutation. A mutation or collection of mutations can make a cell malignant and result in a cancer. Mutations also alter the phenotype, and very often they are either lethal or at least produce unfavorable phenotypes. When the mutations occur in cells producing gametes, they may be passed onto the next generations resulting in variations among offspring.

When such a mismatch is there, the shape of the double helix is distorted. For example, UV radiation can covalently link two adjacent thymine bases, distorting the DNA molecule. One of the two copies of DNA molecules generated by these distorted sequences carries a permanent change in the sequence, a mutation. Such distorted locations are usually recognized and repaired by the DNA repair mechanism of the cell before it becomes permanent, and that reduces the risk of accumulation of mutations. Because DNA repair is very important for the survival of organisms, a large number of DNA repairing enzymes are found in various organisms.

Such enzymes may cut off the mismatched sequences in the damaged strand and replace with correct nucleotides. The cutting (excision) is done by a nuclease and filling the gap, using the correct strand as the template, is done by a type of DNA polymerase. This is called nucleotide excision repair. DNA ligase completes the DNA strand by sealing it with a phosphodiester bond.

Genes and how they work

The nature of prokaryotic and eukaryotic genes

When Gregor Mendel put forward his laws of inheritance in 1860, he used the term hereditary factors to explain what control the characters shown in a phenotype and which are transmitted from generation to generation. At that time, these were imaginary units and where they were located in cellular structure was not known. Today, these physical and functional units of

heredity have been identified as genes that are located on chromosomes as discrete units. This disclosure began with the development of cytology and the ability to observe the behavior of chromosomes in mitosis and meiosis. Behavior of chromosomes and the behavior of the Mendel's hereditary factors show the same pattern. In eukaryotes, chromosomes occur in pairs in diploid somatic cells, and hence genes also occur in pairs. A pair of chromosomes, coming from the two parents, containing the same genes is called homologous chromosomes. Generally prokaryotes have one chromosome in each cell, and hence may be considered haploid.

Gene

Gene is the fundamental physical and functional unit of heredity.

A gene is comprised of a segment of DNA on a specific site on a chromosome. It specifies the sequence of RNA.

The location or the site of a gene on the chromosome is called a locus (loci in plural). The alteration versions of genes that are located at the same locus on different chromosomes are called alleles of the genes. Prokaryotic genes are housed in loci on the circular DNA molecule, as discrete segments of DNA.

There are many steps in a biochemical pathway and each step is controlled by a gene, and therefore many genes are involved in controlling a particular phenotype. In eukaryotes these genes are scattered among several chromosomes, while in prokaryotes, they are arranged in the same region of the chromosome one after the other as clusters. These clusters are expressed together by a single control region and are transcribed into one mRNA. Such mRNA molecules will be translated into several different peptides. These organized gene clusters in prokaryotes are called operons.

Operon: A group of genes that functions as a single transcription unit. It is comprised of a control region (an operator, a promoter) and structural genes that are transcribed into one mRNA. Coding for several peptides.

In prokaryotes, all DNA segments in the chromosome are functional (either transcribed into mRNA or act as control regions), while in eukaryotes, a large proportion of DNA has no identified function. Such DNA segments that are present in between the genes are referred to as intergenic DNA. There are also sequences within the genes that are transcribed, but are not translated into polypeptides. That means there are coding sequences as well as noncoding sequences in the transcript of the gene. The noncoding sequences within the genes are called introns while the sequences that code for polypeptides are exons. Accordingly the transcript contains exons as well as introns. The transcript is a pre-mRNA and it undergoes processing where introns are excised and exons are joined in order to produce mRNA.

Chromosome Theory of Inheritance

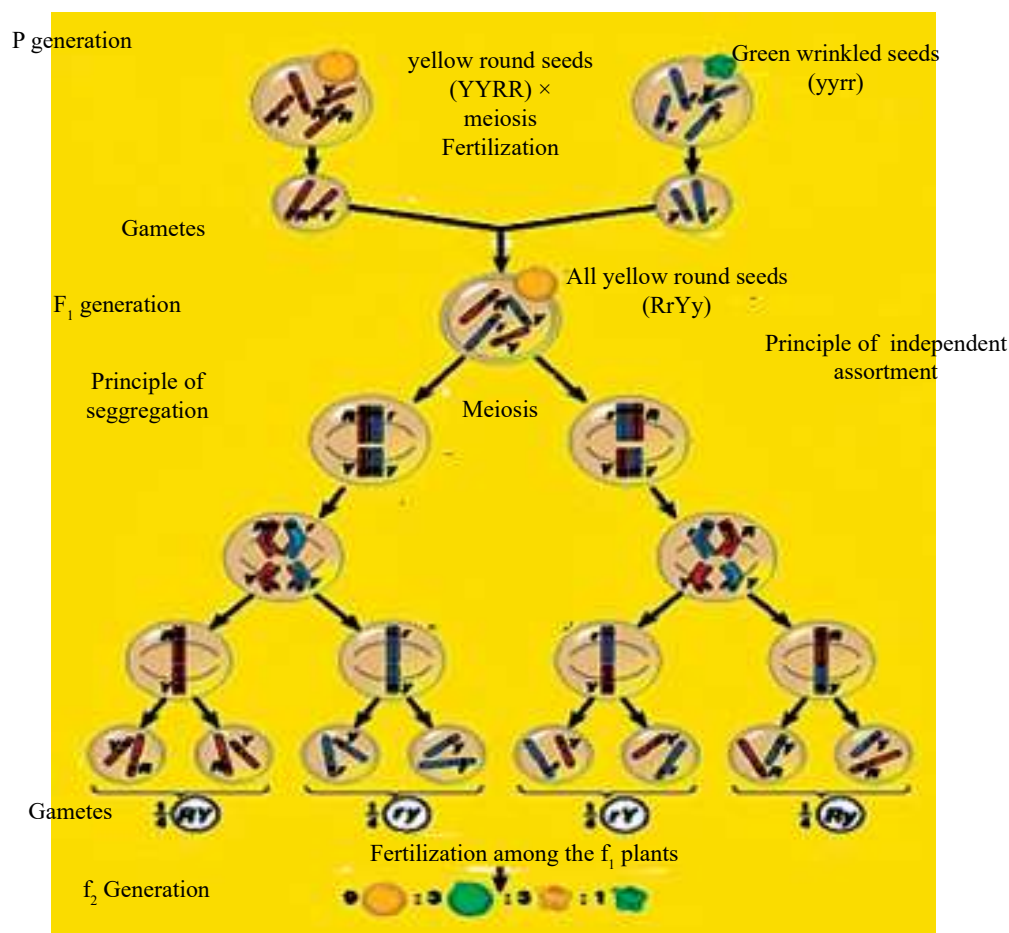


Figure 7.14: Chromosomal basis of Mendel's Laws: How the alleles of genes on homologous chromosomes behave during meiosis

Figure 7.14 shows the parallelism between the behavior of Mendel's hereditary factors and genes and their alleles located on chromosomes. Evidences collected in genetic studies were noted by scientists and several of them independently developed chromosome theory of inheritance. The Mendelian hereditary factors or genes are located on the chromosomes at specific loci. Therefore chromosomes and genes housed on them occur in pairs in diploid cells. The pairing of homologous chromosomes in metaphase I occurs at random and therefore this is independent assortment (i.e. no order in arranging maternal and paternal chromosomes). In anaphase I the independently assorted homologous chromosomes separate reducing the number of chromosomes to half. This is segregation. With the independent assortment and segregation of chromosomes, alleles of genes on nonhomologous chromosomes are assorted independently in different combinations in Metaphase I. The alleles segregate after completion of anaphase I to form four haploid cells having different allele combinations in equal proportions. The same reasoning can explain the phenotypic ratios observed by Mendel in F₂ generation after random crossing of F₁.

Gene Expression

Genes control characters when a gene is at work and, it is said that the gene is expressed. Gene expression is the process by which the information stored in a gene is used to make a functional gene product. The final product of a gene is usually a polypeptide, which after suitable modifications becomes a protein. However several RNAs also function as final gene products. For example, ribosomal RNA (rRNA) and transfer RNA (tRNA) are such RNA molecules, having direct functions.

In examining how genes control characters, it was revealed that (the first suggestion came in 1902 from Archibald Garrod) inherited diseases are caused by the inability to produce related enzymes as a result of inborn errors in metabolism. The symptoms of a hereditary disease condition, known as alkaptonuria is due to inability to make the metabolic enzyme which metabolizes the chemical alkapton. In patients alkapton remains in urea and its oxidation results in black colouration of urine.

The gene expression begins by copying the information stored in a segment of DNA or a gene into sequence of RNA. In synthesis of polypeptide the gene is not directly converted into polypeptide, but involves an RNA molecule to pass the message in DNA to a message in polypeptide. Since the RNA molecule acts as a messenger to communicate the information from DNA to polypeptide, it is called messenger RNA (mRNA). The two steps in polypeptide synthesis are:

The term copying is used in transcription, as in linguistics, since the information written in the nucleotide sequence of DNA by four letters (AGCT) is copied to the same chemical language as a nucleotide sequence of RNA written in four letters (AGCU), with only difference that T is replaced by U. The amino acid sequence in the polypeptide chain is also linear and parallel to the base sequence in the gene or mRNA, but the chemical languages are different: language written in four letters to a language written in 20 letters. This is the reason why the second step is called translation.

1. Transcription - Copying a sequence of DNA into mRNA
2. Translation – Converting the information in mRNA to a sequence of amino acids

Transcription is similar to replication since a DNA strand is acting as a template to make a complementary mRNA strand. The difference in transcription is that the copy is a mRNA molecule and only one DNA strand is copied. The major enzyme catalyzing polymerization is named as RNA polymerase. This is mRNA, because it transmits the message stored in the gene to the place where the polypeptide chain is actually being assembled.

Message in mRNA is translated to a sequence of amino acids. This process occurs in association with ribosomes, which are present in the cytosol.

In addition to mRNA, other types of RNAs and enzymes are involved in the polypeptide synthesis. The basic mechanism of the process of polypeptide synthesis is similar in prokaryotes and eukaryotes, with some important differences.

Genetic Code

In transcription, each letter of the template will be copied to the corresponding letter in mRNA. Note that the mRNA is complementary to template, and hence it is a copy of the other DNA strand. This seems to be straight forward, one to one copying. On the other hand, language of nucleic acid has four letters (nucleotides), whereas the language of protein has 20 letters (amino acids). If each kind of nucleotide is translated into amino acid, only four amino acids would be specified or coded. Therefore a combination of nucleotides would be needed to code for one amino acid. It has been proven by experiment that amino acids are coded by triplets of nucleotide bases and the protein synthesis is based on a triplet code. Therefore, the genetic code is a triplet code. If three-letter combinations or triplets are considered, $4^3 = 64$ possibilities are there. The three letter words or triplets are read one after the other, and hence are non-overlapping. Since all the words have three letters, no space is required to delimit the words. The genetic code stored in a gene in non-overlapping three-letter word is copied to non-overlapping, three-letter word in complementary mRNA. This is translated by reading three letters at a time identifying an amino acid corresponding to each three-letter word (Figure 7.16). The triplet of nucleotide bases of mRNA or coding for an amino acid formation is called a codon. Therefore the genetic code has 64 codons.

Out of 64, 61 triplets code for 20 amino acids, and the other three are used to signal the “stop” of translation or as “termination” codons (UAA, UAG and UGA). AUG codon codes for methionine (Met) as “start codon” signaling the protein synthesizing machinery to start translation of mRNA at that codon. Therefore, the first amino acid of all protein is methionine, but this may be enzymatically removed after the translation. Figure 7.16 shows all 64 codons and what each of them codes for. It is easily noticeable that certain amino acids are coded by more than one codon.

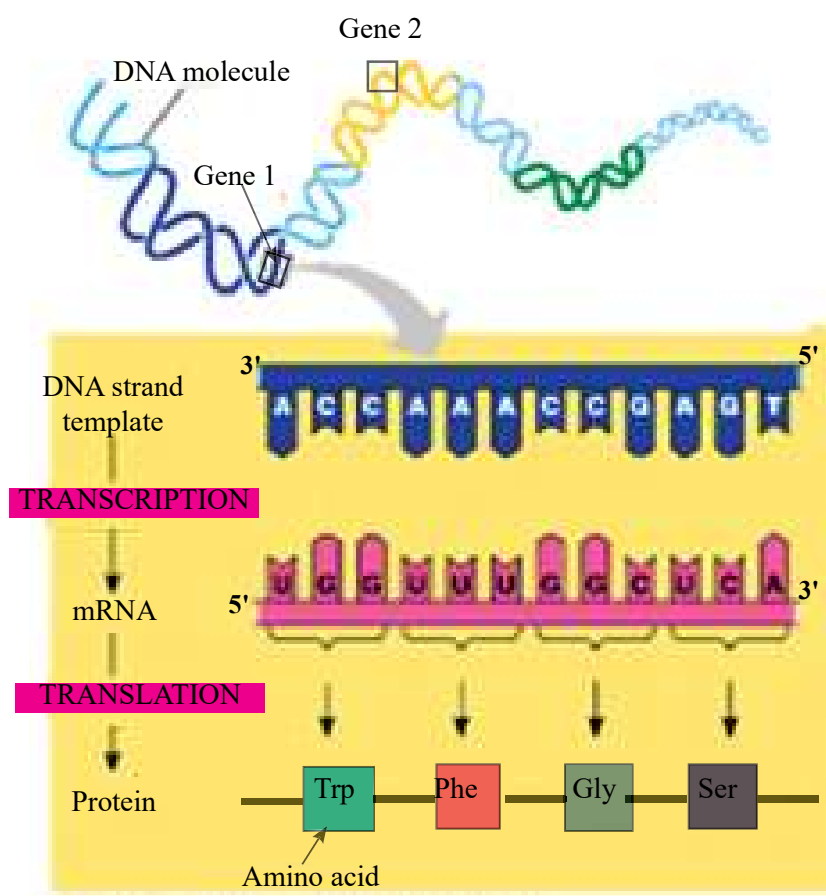


Figure 7.15: The genetic code is copied to mRNA and translated into amino acid in polypeptide chain using triplet codons

To read a message accurately, a starting point and a termination point as well as correct sequence of letters should be identified. This is called a reading frame. The protein synthesizing machinery starts and ends reading at precise positions and reads triplets one after the other in a non-overlapping fashion. Since all words are three letter ones, no spaces between words are required. If the reading starts at a wrong place, entirely wrong message will be read and a wrong polypeptide will be synthesized. If one letter is missing or one letter is added in the reading frame, the wrong message will be read from that point onwards, once again making a wrong polypeptide, if at all. The reading of the message is always from left to right by convention.

Another interesting feature of the genetic code is that its universality. This means nearly all organisms share the same genetic code. Accordingly, a gene isolated from one organism and inserted in to another related or nonrelated organism should express the same protein. This is how human insulin is produced by bacteria. The reading frame for insulin protein will be translated

in exactly the same way in both human and bacterial cells. A firefly gene can be expressed in tobacco plant, so that the plant emits light.

		Second letter					
		U	C	A	G		
First letter	U	UUU } Phe UUC } UUA } Leu UUG }	UCU } UCC } Ser UCA } UCG }	UAU } Tyr UAC } UAA Stop UAG Stop	UGU } Cys UGC } UGA Stop UGG Trp	U C A G	Third letter
	C	CUU } CUC } Leu CUA } CUG }	CCU } CCC } Pro CCA } CCG }	CAU } His CAC } CAA } Gln CAG }	CGU } CGC } Arg CGA } CGG }	U C A G	
	A	AUU } AUC } Ile AUA } AUG Met	ACU } ACC } Thr ACA } ACG }	AAU } Asn AAC } AAA } Lys AAG }	AGU } Ser AGC } AGA } Arg AGG }	U C A G	
	G	GUU } GUC } Val GUA } GUG }	GCU } GCC } Ala GCA } GCG }	GAU } Asp GAC } GAA } Glu GAG }	GGU } GGC } Gly GGA } GGG }	U C A G	

Figure 7.16: The codon table for mRNA

Mechanism of Polypeptide Synthesis

I. Transcription

Transcription is DNA directed RNA synthesis. This is completed in three steps.

1. **Initiation:** The process of transcription is initiated at a specific site called promoter. The promoter site includes the transcription initiation site and several other nucleotides. Only one strand of the double stranded DNA acts as a template for transcription. This is because only the template strand will have the promoter sequence in the correct orientation, which facilitates the binding of RNA polymerase. The enzyme polymerizing the RNA is RNA polymerase. This enzyme binds to the promoter site in correct orientation. The RNA polymerase then unwinds the two DNA strands and begins the transcription at start point. A component of RNA polymerase has the helicase activity, and hence DNA helicase is not involved in transcription.

2. Elongation: RNA polymerase enzymes can start adding complementary ribonucleotides against the template DNA. RNA polymerase continues to add nucleotides in 5' to 3' direction until it reaches the transcription termination site. The DNA strands unwind as the RNA polymerase moves forward, exposing template DNA and allowing pairing with ribonucleotide. The two strands rewind at the other end (figure 7.17)

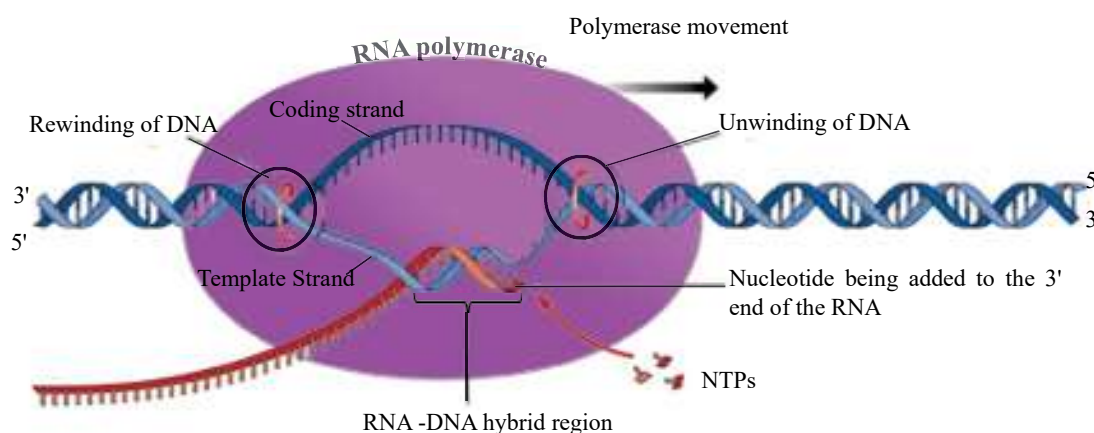


Figure 7.17: Elongation of the newly forming RNA transcript

3. Termination: In prokaryotes, the polymerization continues passing the termination sequence of DNA and RNA polymerase enzyme falls off ending the transcription..
After termination, the newly synthesized pre mRNA in eukaryotes is subjected to RNA processing. The mature RNA leaves the nucleus.

II. Translation

Once the mRNA is in the cytosol, translation process is initiated. Ribosomes read the message written as a sequence of triplet codons in an mRNA and translate it into a sequence of amino acids in a polypeptide with the assistance of transfer RNAs (tRNA). A tRNA attached to the correct amino acid from a pool in the cytosol, transports it to the ribosome which adds the amino acid to the growing end of the polypeptide chain by making a peptide bond. The key players in translation are tRNAs .

A specific tRNA molecule binds a specific amino acid to its one end. Its structure also carries, at a specific location, a triplet of nucleotides, which is complementary to the codon in the mRNA that code for the amino acid it carries. This triplet is known as anticodon and it can base-pair with the codon (figure 7.18). This is how tRNA does the translation, by acting as an adapter molecule between the triplet codon and the amino acid specified by it.

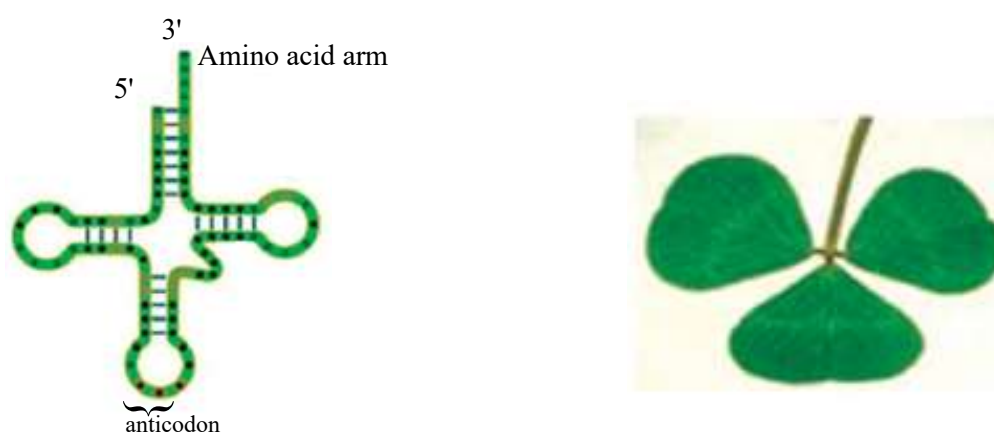


Figure 7.18: Two dimensional model of tRNA showing clover leaf structure

Structure of ribosome

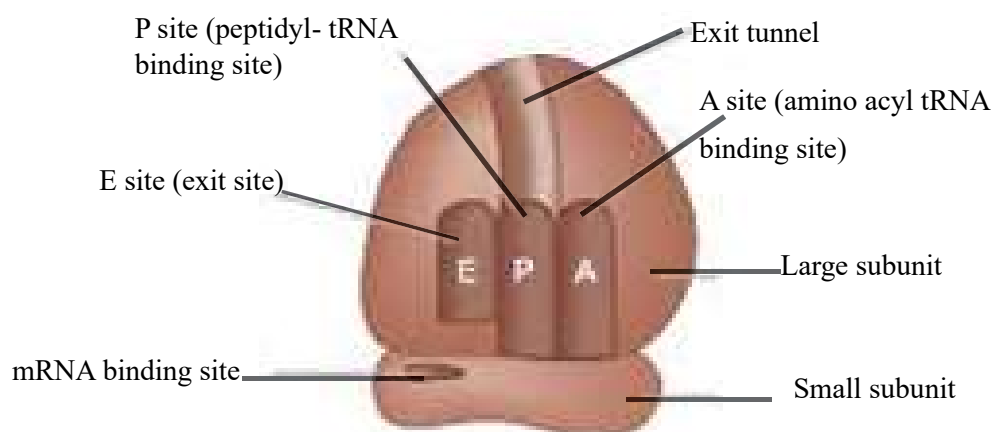


Figure 7.19: Structure of ribosome

Process of translation

The translation is also completed in three stages.

1. **Initiation:** The first step in initiation is binding of small subunit of ribosome to mRNA and to the initiator tRNA which carries methionine as the first amino acid. Then, the two subunits of the ribosome combine to form the functional ribosome. This complex of ribosomal subunits, mRNA and initiator tRNA, is called transcription initiation complex (Figure 7.20). Then the mRNA moves until the AUG start codon, aligns with the P site of the large subunit. Then, the anticodon of the initiator tRNA forms hydrogen bonds with AUG start codon. This signals the initiation of the translation.

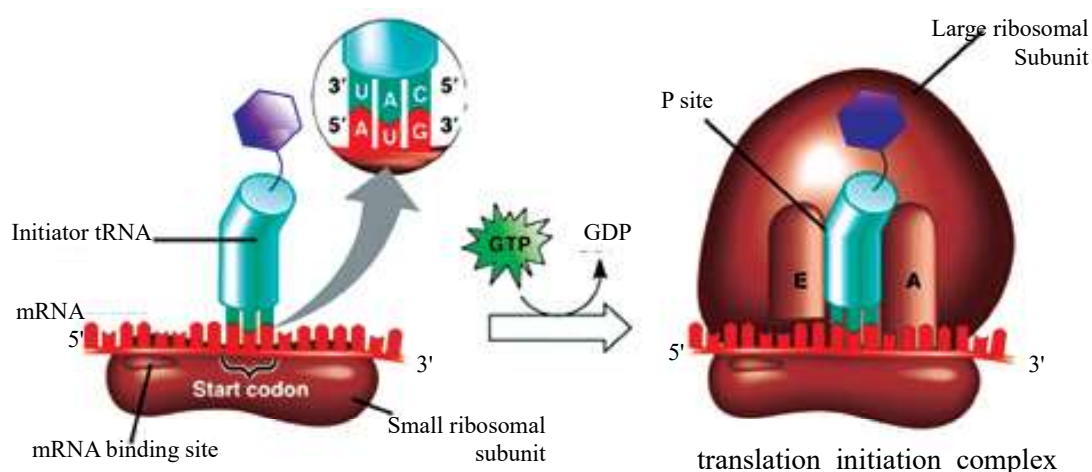


Figure 7.20. Formation of translation initiation complex

2. **Elongation:** In this stage, amino acids are added to the C- terminus of the growing polypeptide chain by peptide bonds, one after the other governed by the triplet codons. Elongation is completed by a three-step cycle. At the end of the initiation stage, P site is occupied with tRNA attached to methionine and A site is empty and is aligned with the next codon. The second tRNA, charged with the corresponding amino acid is brought to the A site, matching the codon and anticodon. This first step in the cycle is the codon recognition. As the second step, a peptide bond is formed between the carboxyl group of the growing polypeptide chain in the P site with the amino group of the amino acid in the A site. An rRNA catalyses this reaction. The third step is the translocation of the mRNA. The mRNA moves codon by codon unidirectionally. In the process, the tRNA with the growing polypeptide chain in the A site is moved to the P site. The released tRNA in the P site will move simultaneously to the E site, from where it is released to the cytosol. The A site is now aligned with the next codon, so that this cyclic process can continue. GTP is used for the energy requirement of the elongation process. See Figure 7.21 for an illustration of elongation stage in translation.

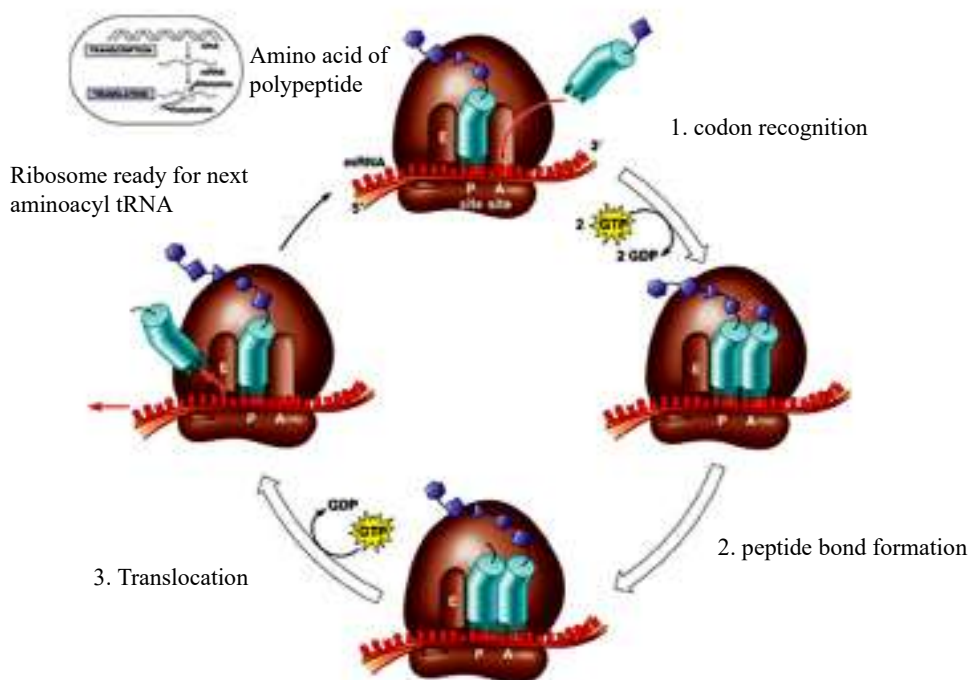


Figure 7.21: Three-steps that occur in cyclic manner in elongation stage in translation.

- 3. Termination:** When the mRNA moves, it finally aligns one of the stop codons: UAG, UAA or UGA with A site. They do not code for any amino acid, hence there is no tRNA which comes into A site. This releases the completed polypeptide chain to the cytosol. The ribosome and the remainder of the translation assemble then fall apart (Figure 7.22).

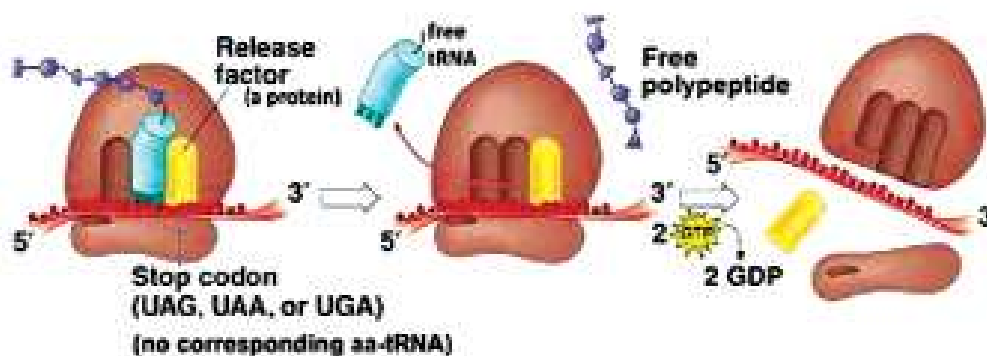


Figure 7.22: Termination of the protein synthesis

Polyribosomes/ Polysomes:

When the mRNA moves away for a sufficient distance, a second ribosome can bind to it. Depending on the length of mRNA, a number of ribosomes can be attached to the mRNA simultaneously. As such, mRNAs that are being translated actively are associated with multiple ribosomes attached to a string of mRNA, forming polyribosomes or polysomes. Forming polyribosomes increases the rate of translation since they allow simultaneous translation by several ribosomes.

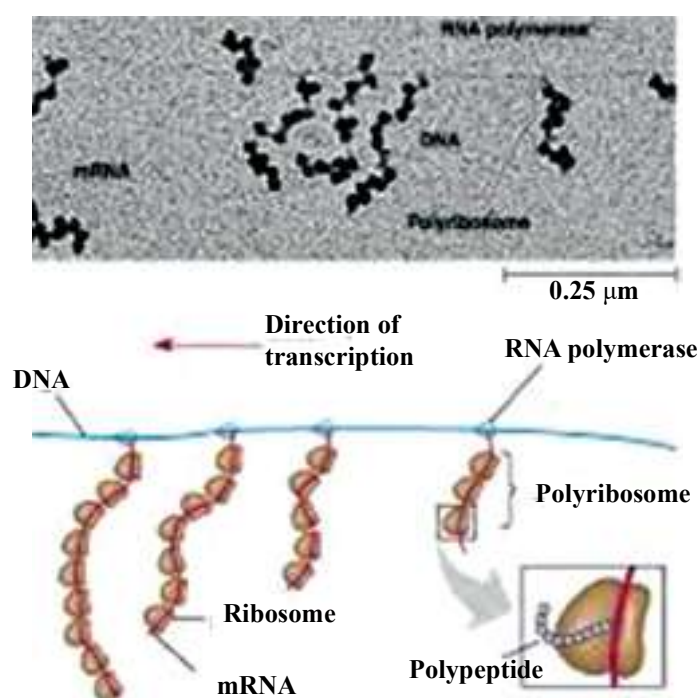


Figure 7.23: Polyribosomes of prokaryotic organism - mRNA is still growing and attached to the DNA

Fate of the proteins

The newly synthesized polypeptide is the primary structure of the polypeptide. This is not functional as a protein as it is. The polypeptide has to assume its functional form by folding (refer unit 02) and sometimes with post-translational modifications.

Certain polypeptides have additional segments than what is required for its function. For example, a short segment of amino acids are present in certain polypeptides to act as signal peptide. Signal peptide guides the polypeptide to a particular location in the cell or to be secreted. This is referred to as protein trafficking. Once the polypeptide is in place, the extra piece of the peptide chain is no longer needed and maybe removed enzymatically.

The post-translational modifications include chemical modification of certain amino acids by attaching sugars (glycoproteins), lipids (lipoproteins), phosphate groups (phosphorylated proteins) and other additions. The first amino acid, methionine, maybe removed enzymatically. Enzymes may also cut the initial polypeptide into two or more pieces which form a functional protein, by joining different combinations. For example, the protein insulin is produced as a single polypeptide, but cut at two places to remove the central piece. The other two polypeptide chains are joined together to form the functional insulin.

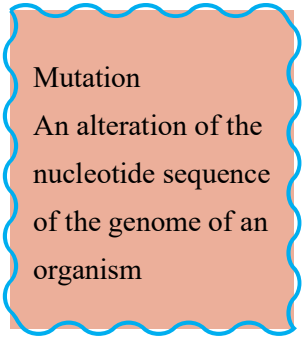
Selective Degradation of Proteins

The amount of a protein in a cell is determined on one hand by the rate of synthesis and on the other by the rate of degradation. The selective degradation of proteins is an essential mechanism in regulation of cellular activities. Certain proteins are degraded in response to specific signals. Faulty or damaged proteins are recognized and rapidly degraded to avoid bad effects due to mistakes in polypeptide synthesis or errors in folding. Some proteins for example regulatory proteins, need to be rapidly degraded after their function. The structural proteins may remain for a longer period.

Mutations

The phenotype of an organism basically depends on the genetic information stored in DNA and the final outcome is a result of the interaction between genetics of the organisms and the effects of the environment.

Alterations in DNA may bring about certain changes in the characters of individuals in a species, which result in phenotypic variation among organisms. These changes occur permanently, and are called mutations.



Mutation
An alteration of the nucleotide sequence of the genome of an organism

Mutations are the sources of variation seen among individuals of a species.

The effect of a mutation may be neutral, beneficial or deleterious. Deleterious mutations may be lethal or at least they are less favourable than the original phenotype. A mutation may also cause complete loss of a function. In rare occasions, the function of a polypeptide may be enhanced due to a mutation. These are beneficial mutations. Entirely new functions may also be resulted by mutations. For example, an enzyme specific to a substrate may change due to mutation in such a way that it changes its specificity to act on another substrate. The product due to the mutation is capable of catalyzing a new biochemical reaction.

There are two major types of mutations based on the scale of alterations in the genetic material; small-scale changes occurring in the sequence of nucleotides in a gene or large-scale changes in the number or the structure of chromosomes. These are known as gene mutations and chromosome aberrations or chromosome mutations, respectively.

Gene mutations

The permanent alterations of DNA sequence of a gene are called gene mutations. They may be occurred due to rare errors occur during DNA replication. These are called spontaneous mutations. In addition, certain external factors can also cause mutations at a higher rate. Since these factors generate mutations, they are called mutagens. Mutagenic agents can be classified as chemical or physical factors. X-rays and UV rays are examples for mutagenic physical agents.

The mutagenic agents can introduce mutations to replicating DNA in a cell. Mutations are also the cause of carcinogenesis. Therefore, mutagens are carcinogens and carcinogens are mutagens. These chemicals and radiations need to be handled with utmost care.

Types of Gene Mutations

These are small-scale mutations involving only one nucleotide pair or more than one nucleotide pair. If only one pair is altered, these are called point mutations. There are three types of gene mutations. They are:

1. A single nucleotide pair substitution – change one nucleotide pair with another
2. Nucleotide pair insertions – addition of one or more nucleotide pairs
3. Nucleotide pair deletions – removal of one or more nucleotide pairs

The substitution of a pair of nucleotide is a point mutation. Insertions or deletions can be point mutations, or may involve more than one nucleotide pair.

Substitution

Here one nucleotide pair is replaced by another pair (Figure 7.23). There is no change in the length of the gene. Because the same amino acid may be coded by more than one codon. Some substitutions are silent mutations. Substitution of one nucleotide pair of a gene may not have an effect on the polypeptide it codes for. Because the same amino acid may be coded by more than one codons. The third letter in the codon triplet has a wobble, meaning that even if the third letter of a codon is replaced by another letter the same amino acid is coded in the third letter (for example, the codon 5'-GGC-3' on the mRNA will be modified to 5'-GGU-3', if 3'-CCG-5' triplet on the DNA template strand be altered as 3'-CCA-5' by substitution G with A). A substitution may also change one amino acid in the polypeptide. Therefore, the meaning of the primary structure of the polypeptide is changed slightly; hence, these mutations are missense mutations. This substitution of the amino acid with another amino acid may or may not have significant impact on the tertiary or quaternary structure (functional form) of the protein. Occasionally, this may render the protein a higher activity or even with new properties. However, very often, these changes are either neutral or detrimental, making the protein useless or less efficient.

It is also, possible that a point mutation converts a codon coding for an amino acid to a stop codon. This causes premature termination of the protein synthesis, and therefore it is called non-sense mutation (figure 7.23). The result is a shorter polypeptide than original, which is usually non functional.

Insertion and deletion

These mutations bring about drastic changes in polypeptides compared to substitutions (note: nonsense mutations due to substitutions may also result in major changes). Insertion or deletion of a nucleotide or a couple of nucleotides will result in shift in the reading frame, reading wrong codons after the point of mutation. Therefore, such mutations are called frameshift mutations (figure 7.23), resulting in extensive missense. The polypeptide may not be functional at all, unless the insertion or deletion occurred is very close to the termination codon. It may also introduce a new stop codon, which was not in the original sequence. If that is the case, the translation terminates causing nonsense mutation. However, if the insertion or deletion is a triplet or a multiple triplet, the reading frame, will be back to the original reading frame immediately after the point mutation (Figure 7.23). In such cases, one or a small number of amino acids will be added or removed, respectively from the entire sequence. The message will only be slightly changed and the polypeptide may be functional, depending on the importance of the mutated region of the polypeptide in its correct folding.

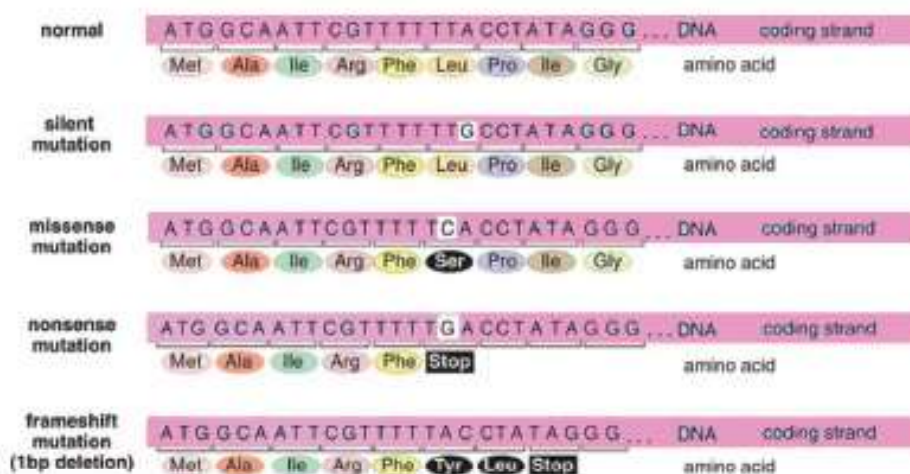


Figure 7.23: Types of Gene mutations

Chromosome Aberrations/ Chromosome Mutations

Since many genes are involved, most of the chromosomal mutations are lethal, while others are deleterious. Spontaneous abortions or miscarriages in mammals occur due to unusual numbers or structures of chromosomes. Such mutations also bring about various developmental disorders. Beneficial chromosome mutations are extremely rare. In plants, some chromosomal mutations cause beneficial variations.

I. Mutations due to alterations in chromosome structure:

Large chunks of chromosomes, which may include several genes to several hundred genes, may be lost, moved to another chromosome (cut and paste), copied and moved to another chromosome (copy and paste), or the orientation changed in chromosome mutations. These four types of chromosome mutations are called deletion, translocation, duplication and inversion (Figure 7.24).

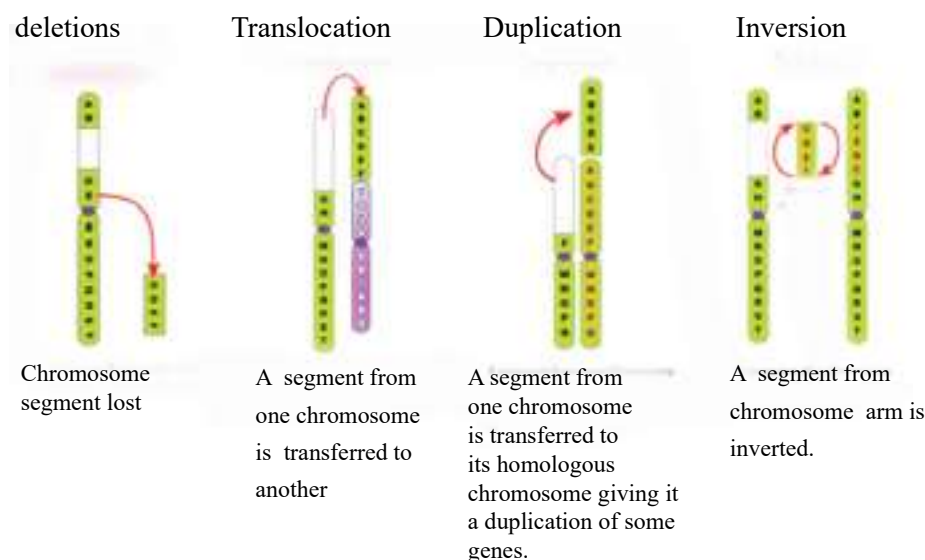


Figure 7.24: Four types of chromosomal mutations

The loss of a part of a chromosome removes several genes, and hence often these changes are lethal. In translocation, there is no loss of total amount of DNA. However, the expression of genes can be altered due to the change in the environment at the new location. The cutting of the chromosome may occur within a gene, and if this happens, that gene cannot function. In duplication, an extra piece of DNA carrying many genes is present in another location of the genome. This situation may also alter the gene expression, usually leading to deleterious impacts on the phenotype. The change of the orientation of a chunk of chromosome or inversions can also result in changes in gene expression; most of these are adverse variations.

II. Mutations due to alterations in chromosome number:

In addition to the structural changes of the chromosomes, a cell may contain an entire chromosome or even a set of chromosomes in excess to the normal number of chromosomes. A cell may also get one chromosome less than the normal number. When there is one more or one less chromosome in a cell, that situation is referred to as aneuploidy. Here the ploidy level has not changed, but when a complete set of chromosomes are present in excess, then ploidy level is said to be increased, for example to triploid, tetraploid, hexaploid etc.

Aneuploidy is resulted in by the mistakes in meiosis. During meiosis I, the two sets of chromosomes in a diploid cell must separate and move towards the two poles of the cell. However, due to abnormal arrangement of homologous chromosomes, both chromosomes of a pair may migrate to one pole, so the other end has one less chromosomes. The resulting cells or gametes in sexual reproduction also have either one more or one less chromosomes than the haploid number of chromosomes. A similar result is obtained when chromatids of a chromosome do not separate and migrate to the opposite poles of the cell in meiosis II. The inability to separate a pair or pairs of chromosomes in meiosis is called nondysjunction (Figure 7.25). When a gamete with one less chromosomes, unites with a normal gamete, the resulting zygote is an aneuploid with $2n-1$ chromosomes. This type of cells is called monosomic, since a specific chromosome has only one of it. A gamete with one more than the usual haploid set of chromosomes may fuse with a normal gamete. The zygote will carry a chromosome in triplicate, and hence $2n+1$. This aneuploidy status is called trisomic for that chromosome.

Such abnormalities can take place also in mitosis. Ploidy level can also increase by abnormal separation of chromosomes. A triploid ($3n$) can result in fertilization of an abnormal diploid egg. If a zygote does not divide after the first mitotic division, it will carry four sets of chromosomes and be developed into a tetraploid ($4n$).

Animals with higher ploidy level are very rare. Plants on the other hand, can tolerate higher ploidy levels, and often perform better than their diploid individuals.

Examples of plants having higher ploidy levels:

Banana – Triploid ($3n$); Wheat – hexaploid ($6n$); Strawberries – Octaploid ($8n$)

Polyploids are more common in invertebrates than in vertebrates. Among vertebrates, polyploidy is observed in a few fishes and amphibians.

Polyploids are rather normal compared to aneuploids. Although having higher number of chromosomes than usual, polyploids maintain the genetic balance, whereas genetic balance is lost in aneuploids.

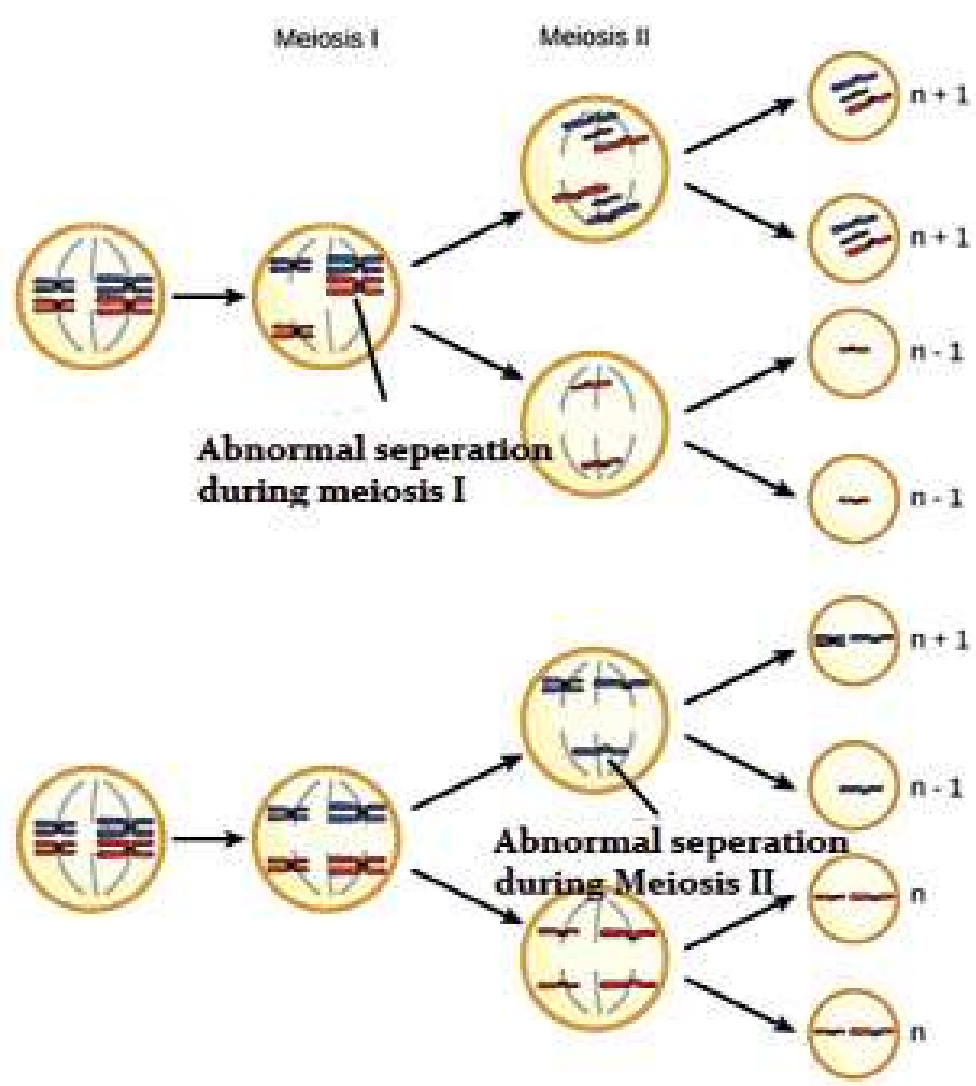


Figure 7.25: Abnormalities in meiotic division causing aneuploidy

Human Genetic Disorders

I. due to Gene Mutations

Two examples of human genetic disorders by gene mutations are described below.

Colour Blindness

Colour blindness or Colour Vision Deficiency is a genetic disorder, more common in males than in females, due to mutations on one or more genes located in X chromosome. These genes code for proteins that absorb different wavelengths of visible light. These are visual pigments called photopsins, which are categorized as red, green or blue. A person with a normal colour vision can distinguish different colours and their hues because they have all three categories of pigments in the retina. The absorption of different proportions of different wavelengths by the pigments is resolved by the brain as the colour of the object. In human genes coding red and green pigments are located in X chromosomes and gene for blue pigment is on the chromosome 7. Since males have only one X chromosome and the relevant genes are absent in Y chromosome, any defect in one or both of these genes will make the phenotype. In females at heterozygous condition a defective allele in one X chromosome is masked by the good allele in the other. Therefore colour vision deficiency is more common in males (5-8% of males) than in females (less than 1% of females). Colour blindness nearly always affects the perception of red or green colours, because these are the genes that are sex linked.

Sickle Cell Aneamia

Sickle cell aneamia is a genetic disease prevalent in human populations in Africa and other warmer regions of the world. A mutated allele of a gene coding for β -globin subunit of hemoglobin, the oxygen carrying pigment, causes an abnormality of the hemoglobin molecules. The presence of this abnormal hemoglobin in red blood cells changes the shape of the RBCs to curve as a sickle from its disk-shape. Individuals with this disorder have fewer numbers of RBCs, and hence develop aneamia. This is because the sickled RBCs breakdown prematurely. The mutation substitutes glutamic acid at a particular place in the primary structure of the β globin with valine (Figure 7.26), which results in abnormal folding of hemoglobin. The mutated allele is codominant, which means both normal β globin and mutated β globin are produced in individuals who are the heterozygous for this locus. Therefore, they have both good and bad hemoglobin and hence both normal and sickled RBCs are present. They are normally healthy and are the carriers of the mutant allele. Since the mutated alleles causes severely determintal effects in homozygous individuals, normally they should have been eliminated from the human population by natural selection. However, there is prevelence of malaria in warmer countries like in Africa, the heterozygous individuals will survive malaria attacks better than individuals with homozygous wild type alleles. This is because, the malaria parasite can not survive in sickled RBCs. Therefore, at heterozygous individuals, parasite density remains at a lower level.

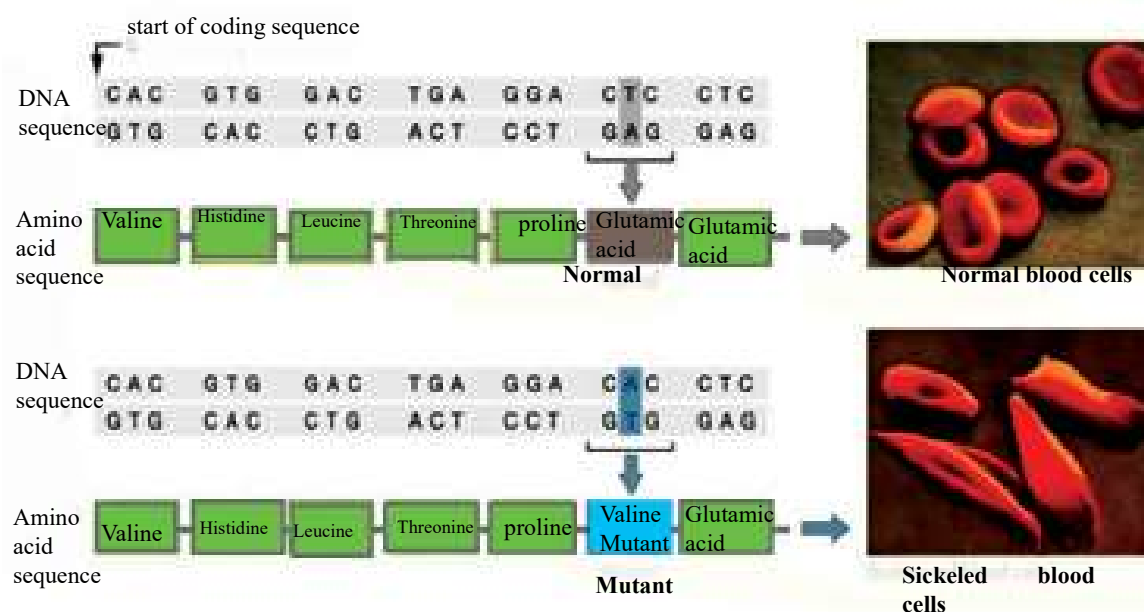


Figure 7.26: The molecular basis of sickle cell anaemia

II. Due to Chromosomal Mutations

Chromosomal mutations bring about drastic changes in amount of genetic material or chromosome structure, leading to abortions of the fetus in mammals. If they survive, they show a peculiar set of abnormal characteristics in the phenotype which are called syndromes.

Three genetic disorders of humans due to aneuploidy are described below.

Down syndrome:

Down syndrome is also known as trisomy -21, because the cells of the affected individual contain an extra copy of chromosome 21. The syndrome shows characteristic facial features, short body, heart defects (which can be corrected) and developmental delays. They have a high risk of developing leukemia and Alzheimer disease. Almost all males and half of the females with Down syndrome are sexually under developed and sterile. Their life span is shorter than normal, but can live past middle age with proper medical treatment. However, they have a lower risk of high blood pressure, arteriosclerosis (hardening of the arteries), strokes and many types of solid tumors. Despite their abnormalities, many live independently and are employed. The risk of having a baby with Down syndrome increases with the age of the mother. This is caused by nondisjunction in meiosis-I. While Down syndrome is due to trisomy of an autosome, there are human genetic disorders due to aneuploidy in sex chromosomes as well. The aneuploidy conditions in sex chromosomes cause Turner syndrome due to monosomy and Klinefelter syndrome due to trisomy.

Turner syndrome:

Turner syndrome is due to monosomy in X chromosome. In very rare cases, there are females with only one X chromosome, and hence their genotype is XO. This is the only known viable monosomy in humans. These individuals are phenotypically female, but they are sterile because their sex organs do not mature. When girls affected with Turner syndrome are treated with estrogen replacement therapy, they develop secondary sex characters.

They have a short stature and some may have an extra skin on the neck (webbed neck). Other characters are puffiness or swelling (lymphedema) of the hands and feet, skeletal abnormalities, heart defects, high blood pressure and kidney problems. Most of them have normal intelligence.

Klinefelter syndrome:

This is due to rare condition having an extra X chromosome in the genotype of XXY. Since it carries a Y chromosome, these individuals are males. Although these people have male sex organs, they are sterile. Their testes are abnormally small. One out of the two X chromosomes is inactivated. Yet these males may have enlarged breasts and may develop other female body characteristics. They have subnormal (less than normal) intelligence.

XXY trisomy will make males and XXX trisomy will make female in females. They do not show any syndrome and have normal male and female characteristics, respectively. They are fertile and slightly taller than the average.

Genetic Counseling

Genetic counseling is a service which is important for families having genetic disorders or families having a risk of genetic disorders. This service is intended to estimate the risk a couple may have in conceiving a child with genetic disorders and providing necessary advice to avoid such instances. Genetic counseling is a profession that requires, on one side, a good knowledge of human genetics to understand, how traits obeying the rules of simple Mendelian inheritance would behave and on the other side, an ability to provide guidance to minimize the risk of having a child with a genetic disorder. If the family already has such a child, the genetic counselor would also provide advice how to manage the situation and how to plan the birth of next child.

Some of the genetic disorders are multifactorial, which means several factors including multigenic inheritance and that is also affected by environment. For example, heart attacks and diabetes may be inherited, but the risk of developing the disease is affected by external environmental factors such as life style and food habits. As such, a clear pattern of inheritance of the disease cannot be traced. The risk of the effect of a trait that follows simple Mendelian laws of inheritance on a child to be conceived can be estimated by studying the history of the family with regard to a

particular disorder, and hence, becomes the subject of genetic counseling. If the disorder is caused by a dominant allele, it can easily be observed in the potential parents. However, if the allele is recessive, a parent or both the parents with normal phenotype may be homozygous for dominant allele or heterozygous carrier. Tracing the history of the family with regards to the disease using pedigree analysis, would allow estimation of probability of parents being carriers and accordingly would estimate the probability of the risk of having a child with the disorder. The information available through pedigree analysis, sometimes is sufficient to determine exactly the genotype of one or both parents. The genetic counselor explains the situation to the potential parents and guides them to choose the most suitable option in having a child.

Techniques are now available to determine whether the already conceived fetus carries the mutant alleles. For this, the cells of the early fetus are sampled and DNA is sequenced to find out the presence or absence of the mutant allele and, if present, whether the fetus is homozygous or heterozygous. This information is very important to take well informed decision to keep the fetus or to abort it. Legislations of some countries allow abortions of such fetuses which would otherwise be born with genetic disorders. This however is a difficult decision for parents to make. Therefore genetic counselor has a task of guiding the potential parents to make the best decision possible.

Gene Technology

Tools, Techniques and Methods in Gene Technology

This section examines the process of gene technology starting from DNA isolation, through identification of desired DNA sequences to gene technology or recombinant DNA technology. The isolated DNA needs to be cut different pieces of DNA must be joined and at times DNA must be copied *in vitro*. These involve several enzymes acting on DNA. Distinguishing a unique DNA sequence from the rest of the DNA, requires separation of DNA based on the size of the fragments and identification of it. In making genetically modified organisms, this DNA needs to be transferred to a recipient organism using a suitable method. Copying of DNA is possible *in vivo* using cloning and *in vitro* using Polymerase Chain Reaction (PCR). Sequencing of DNA has become a very important technique in many studies in DNA.

Isolation of DNA

Gene technology begins with isolation of the target DNA sequence from the total genome of the donor cells. Purified DNA is required for many applications such as studying DNA structure and chemistry, examining DNA-protein interactions, carrying out DNA hybridizations, sequencing, PCR, performing various genetic studies or gene cloning.

Since DNA molecules are very long, the entire length of a DNA molecule cannot be isolated, except for shorter DNA molecules such as plasmid DNA and viral DNA. However, breaking of DNA or shearing has to be minimized during the extraction process.

The basic principles and major steps of DNA isolation can be identified as follows.

- **Homogenization or disruption of cells:** DNA is located in the nucleus of the eukaryotic cells and it is concentrated in nucleoid in prokaryotic cells. The first step in DNA isolation is releasing the DNA from the cells by breaking cells or lysis of the cells. Cells can be lysed by mechanical methods such as grinding and homogenization or enzymatically by lysozyme to break bacterial cell wall.
- **Inhibition of DNase:** When the cells are broken, the DNA may get in touch with DNA degrading enzymes such as Deoxyribonuclease (DNase). Therefore, DNA must be protected from such enzymes causing shearing. Chelating agents are added to remove metal ions required for nuclease activity.
- **Dissociation of nucleoprotein complexes:** DNA needs to be freed from the proteins that bind to DNA. DNA-protein interactions are disrupted with SDS, phenol, or proteolytic enzymes.
- **Removal of contaminating materials:** All other molecules in a cell are contaminants of DNA. Removal of this contaminants is required for some applications.
- **Precipitation of DNA:** Here the DNA dissolved in aqueous phase is precipitated with cold (0 °C) ethanol. The precipitate is usually re-dissolved in a buffer. RNA is removed by limited treatment with DNase free RNAase (ribonuclease).

Enzymes Reacting with DNA

Enzymes are required in cutting, joining and copying of DNA *in vitro*.

1. **Restriction endonucleases:** There are several different types of nucleases in cells having different functions. In gene technology, cutting DNA at precise locations is important. The enzymes that recognize specific sequences of DNA and cuts at or near these sites are called restriction endonucleases. The position where the DNA sequence is cut is called as restriction site or cleavage site (Figure 7.28).
eg. *EcoRI*- Source: *E. coli*
2. **DNA Ligase:** Cut DNA fragments from different sources are joined together by forming a phosphodiester bond by DNA ligase to obtain recombinant DNA molecules (Figure 7.27). T4 DNA ligase is the most commonly used DNA joining enzyme in gene technology. The source of this enzyme is T4 bacteriophage.

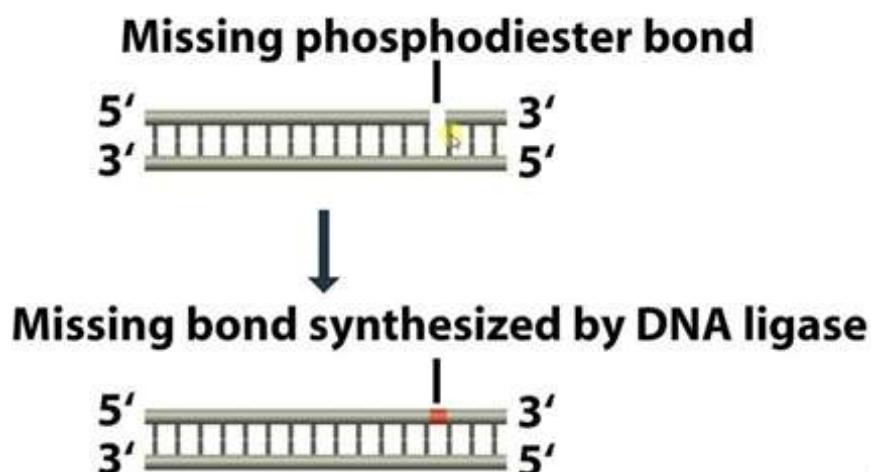


Figure 7.27: Formation of a new phosphodiester to fill the gap between adjacent nucleotides

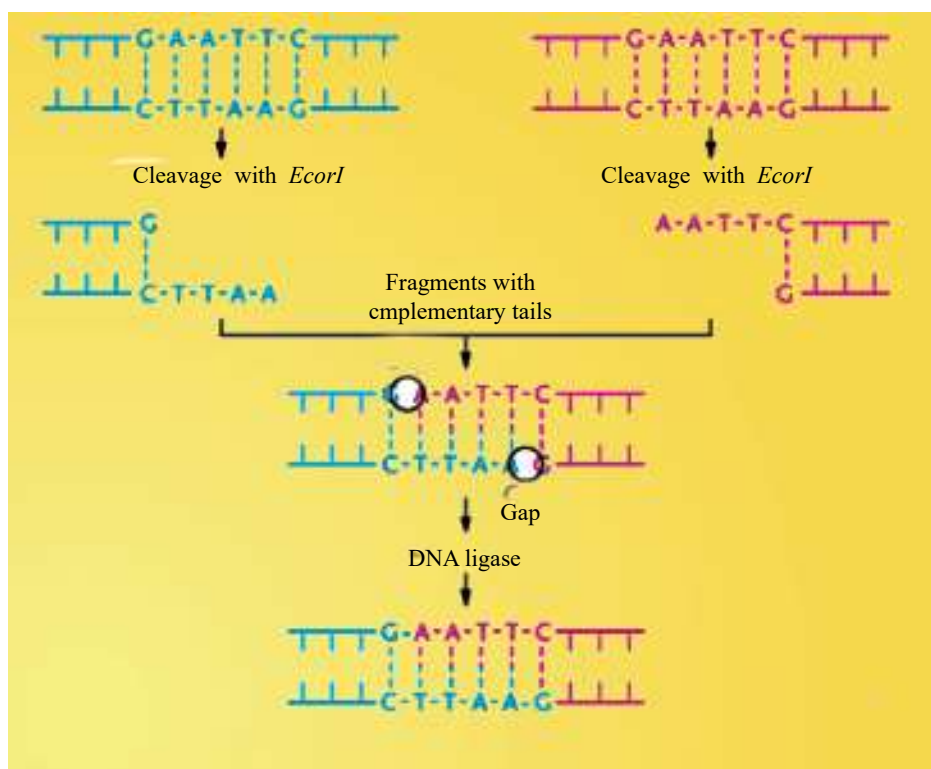


Figure 7.28. Cutting of DNA of different origins with *EcoRI* restriction enzyme and joining fragments of the different DNA by DNA ligase to form a recombinant DNA molecule.

3. **DNA Polymerases:** These are enzymes that add complementary deoxyribonucleotides to the template strand in a growing DNA strand, so that they copy DNA. These are therefore very important in gene technology especially in PCR and DNA sequencing. The most widely used DNA polymerase is Taq DNA polymerase. This is a heat stable enzyme isolated originally from the thermophilic bacterium *Thermus aquaticus*.